



Centre for Advanced Studies

Dr. A.P.J. Abdul Kalam Technical University Uttar Pradesh,
Lucknow

w.e.f.

Session 2022-23

Syllabus and structure for

M. Tech.

In

Manufacturing Technology & Automation



CENTRE FOR ADVANCED STUDIES
Dr. APJ Abdul Kalam Technical University, Lucknow

M.TECH – MANUFACTURING TECHNOLOGY & AUTOMATION

Effective from 2022-23

Program Structure

The Centre for Advanced Studies (Dr. A.P.J. Abdul Kalam Technical University, Lucknow, U.P.) is starting Master of Technology in Manufacturing Technology and Automation from the academic session 2019-20. The curriculum has been developed considering the present and future needs of industry and higher education. The Centre will facilitate both industry ready and research based ambience to students with world class e-library, renowned faculty members to achieve academic excellence and other services.

This is a program aimed to provide the student with in-depth knowledge of domains in the field of Manufacturing Technology and Automation.

M. Tech in Manufacturing Technology and Automation. It is a discipline dealing with the integration of mechanical, manufacturing, automation, measurements and smart manufacturing. It is unique balance of rigorous fundamentals and engaging real-world applications in Manufacturing Engineering which provide the students the opportunity to design, build, operate and maintain machinery equipped with mechanical, electronic/electrical and computer control systems.

The course structure will help students to develop knowledge and skill in the following proportionate:

90% Technical /Research	Precision Engineering, Theory of Machining, Thermo Production Processes, Fundamentals of Tribology, Additive Manufacturing, Kinematics & Dynamics of Robots, Finite Element Method, Mechanical Vibration, Laser Processing of Materials, Advanced Manufacturing Processes, Computer Aided Manufacturing, Theory of Metal Forming, Advanced manufacturing system, Composite Materials, Optimization Theory, Micro-Electro-Mechanical-
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	Systems (MEMS), Automation and Control, Quality Engineering and Management
10% Business Process & Managerial Practices	Know the Business Dynamics, Business Processes and good planning, ability to work with all management level and resolve issues, business need with manufacturing and automation requirement. Consulting Skill, Communication Skill, Legal Understanding.

The students of Manufacturing Technology and Automation programme will attain following objectives:

- a) Students will be able to understand the evolving Manufacturing Technology and Automation from their underlying physical principles and properties.
- b) Students will be able to design and develop Manufacturing Technology and Automation systems by applying underlying mathematical principles, supporting software for engineering model preparation and their analysis.
- c) Students will be proficient in the systematic explorations of alternatives for Manufacturing Technology and Automation.
- d) Students will be proficient in the use of communications (oral presentations and written reports) to articulate their ideas effectively.
- e) Students will be prepared for the continuing learning and self-improvement necessary for a productive career in Manufacturing Technology and Automation.

Program Structure:

M. Tech course is a full time two-year program. The Program Structure has been designed such that the students shall study core subjects and elective subjects of Manufacturing Technology and Automation.

In Semester 1, students shall study fundamental core courses related to Manufacturing Technology and Automation while in the next two semesters students shall study in depth Advanced core and elective subjects. Semester IV is Thesis/ Dissertation.

Every core course consists of lecture (L) hours, tutorial (T) hours and practical (P) hours. Elective courses consist of Lecture (L) hours only. The credit for a course is dependent on the number of hours of instruction per week in that course, as given below:

- (1) 1h/week of Lecture (L) = 1 credit
- (2) 2h/week of Practical (P) = 1 credit
- (3) 1h/week of Tutorial (T) = no credit
- (4) Credit for a Theory course = No. of hours of lectures per week = L
- (5) Credit for a Lab course = $\frac{1}{2}$ * No. of hours of lab per week = P

Course Credit Distribution

Category	Core Courses	Elective Courses	Labs	Project	Total Credit
Credits	24	9	5	22	60

SEMESTER	COURSE TYPES	CREDITS	
	Semester I		
1	Core Course	(3 credit * 4 course)=12	TOTAL COURSE =5
1	Elective Course	(3 credit * 1 course) = 3	TOTAL SEM CREDIT = 18
1	Core Course Lab and seminar	1 credit * 3 course = 3	
	Semester II		
2	Core Courses	3 credit * 3 course = 9	TOTAL COURSE =5
2	Elective Course	3 credit * 2 course = 6	TOTAL SEM CREDIT = 17
2	Core Course Lab and comprehensive viva-voce	1 credit * 2 course = 2	
	Semester III		
3	Core Courses	3 credit * 1 course = 3	TOTAL COURSE =1
3	Thesis - I	9 credit	TOTAL COURSE =CREDIT = 12
	Semester IV		
4	Thesis - II	13 credit	MAJOR PROJECT TOTAL SEM CREDIT = 13
		TOTAL PROGRAM CREDIT	60

MANUFACTURING TECHNOLOGY AND AUTOMATION

PROGRAM

Programme Educational Objectives (PEOs)

- PEO1.** Students will develop into independent researchers and academicians in the broad area of Manufacturing Technology and Automation areas.
- PEO2.** Students will demonstrate a high level of competency and problem-solving aptitude to find innovative solutions for theoretical and practical problems.
- PEO3.** Students will demonstrate active involvement and participation in Research and Development leading to innovative contributions at both national and international levels.
- PEO4.** Developing a practice of continuously updating with latest knowledge and information in their relevant field of specialization.
- PEO5.** Students should engage with engineering profession and understand the importance of ethics, team work and professionalism.

PROGRAM OUTCOMES

1. **PO1:** An ability to independently carry out research/investigation and development work to solve practical problems.
2. **PO2:** An ability to write and present a substantial technical report/document.
3. **PO3:** Students should be able to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
4. **PSO 1:** An ability to apply knowledge and skill of various latest approaches in manufacturing technology and automation. To learn a skill to automate a mechanical system or a process to meet desired needs within realistic constraints such as health, safety and manufacturability.
5. **PSO 2:** The research based knowledge and research methods including design of experiments, analysis and interpretation of data and IT tools in manufacturing.

SEMESTER –I

S No.	Course Code	Subject	Periods (Hrs/week)			Credits
			L	T	P	
1	AMTA-101	Advanced Engineering Mathematics	3	0	0	3
2	AMTA -102	Metal removal processes	3	0	0	3
3	AMTA -103	Materials Processing	3	0	0	3
4	AMTA -104	Sensors and Actuators	3	0	0	3
5		Elective- I	3	0	0	3
6	AMTA -151	Lab-I Sensors, Drives and Control lab	0	0	2	1
7	AMTA -152	Lab-II Manufacturing-Lab	0	0	2	1
8	AMTA -153	Independent Study and Research (ISR) - I	0	0	2	1
		TOTAL	15	0	6	18

SEMESTER –II

S No.	Course Code	Subject	Periods (Hrs/week)			Credits
			L	T	P	
1	AMTA -201	Fundamental of Robotics & Automation	3	0	0	3
2	AMTA -202	Micro Manufacturing Processes	3	0	0	3
3	AMTA -203	Advanced Characterization Techniques	3	0	0	3
4		Elective- II	3	0	0	3
5		Elective- III	3	0	0	3
6	AMTA -251	Robotics and Automation lab	0	0	2	1
7	AMTA -252	Independent Study and Research (ISR) - II	0	0	2	1
		TOTAL	15	0	4	17

SEMESTER –III

S No.	Course Code	Subject	Periods (Hrs/week)			Credits
			L	T	P	
1	AMTA -301	Research Methodology and Technical Writing	3	0	0	3
2	AMTA -352	Thesis – I	0	0	14	9
		TOTAL	3	0	18	12

SEMESTER –IV

S No.	Course Code	Subject	Periods (Hrs/week)			Credits
			L	T	P	
1	AMTA -451	Thesis– II	0	0	26	13
		TOTAL	0	0	26	13

Elective Subjects-

Elective- (I)

S. No.	Course code	Subject
1	AMTA -111	Friction, wear and lubrication
2	AMTA -112	Finite Element Method
3	AMTA -113	Precision Engineering
4	AMTA -114	Composite Materials
5	AMTA -115	Automation and Control
6	AMTA -116	Thermo Production Processes

Elective- (II-III)

S. No.	Cluster	Course code	Subject
1	A	AMTA -211	Additive Manufacturing
2		AMTA -212	Theory of Metal Forming
3		AMTA -213	Machine tool control and condition monitoring
4		AMTA -214	Laser Processing of Materials
5		AMTA -224	Modelling and Simulation
6		AMTA - 226	Drone Technology
7	B	AMTA -211	Additive Manufacturing
8		AMTA -215	Biomaterials Processing and Applications
9		AMTA -216	Surface Engineering
10		AMTA -217	Modelling and Simulation of Materials
11		AMTA -218	Optimization Theory
12		AMTA -219	MEMS and NEMS
13		AMTA -224	Modelling and Simulation
14		AMTA - 226	Drone Technology
15	C	AMTA -211	Additive Manufacturing
16		AMTA -220	Digital Manufacturing
17		AMTA -221	Computer Aided Manufacturing
18		AMTA -222	Inspection and Testing in Manufacturing
19		AMTA -223	Flexible Manufacturing System
20		AMTA -224	Modelling and Simulation
21		AMTA -225	Computational fluid dynamics
22		AMTA - 226	Drone Technology

Labs

S No.	Lab code	Lab Name
1	AMTA -151	Sensors, Drives and Control lab
2	AMTA -152	Manufacturing-Lab
3	AMTA -153	Independent Study and Research (ISR) - I
4	AMTA -251	Robotics and Automation lab
5	AMTA -252	Independent Study and Research (ISR) - II
6	AMTA -351	Thesis - I
7	AMTA -451	Thesis - II

CENTRE FOR ADVANCED STUDIES

Dr. APJ Abdul Kalam Technical University, Lucknow

SYLLABUS FOR M. TECH (Manufacturing Technology and Automation)

Effective from 2022-23

INDEX

S. No.	Title	Page No.
1	Semester- I	13
2	Semester- II	38
3	Semester- III	79
4	Semester- IV	82

MARKS DISTRIBUTION

Following is marks distribution for courses depending upon the credit of the courses:

(1 credit = 50 marks)

Credit	Type of course	Max. Marks	Major (End semester examination)	Performance evaluation		Lab evaluation
				Minor I & II	Internal Assessments (Attendance, Quiz, Seminar & Assignments)	
1	Lab course	50	25	25		-
2	Theory course	100	50	30 (15 each)	20	-
3	Theory course	150	75	40 (20 each)	35	-
4	Theory course with lab	200	75	40 (20 each)	35	50
4	Thesis -I	450	300	150		-
5	Thesis -II	650	400	250		-

Note: Maximum Marks is fixed for all the courses. Only internal assessments and lab evaluation marks distribution may vary as per the instructions communicated by the assigned Faculty. Refer to M. Tech Ordinance & Regulation for Assessment Procedures and Performance Evaluation.

SEMESTER-I

CORE COURSES

AMTA-101	ADVANCED ENGINEERING MATHEMATICS	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Linear Algebra: Matrix algebra; basis, dimension and fundamental subspaces; solvability of $Ax = b$ by direct Methods; orthogonality and QR transformation; eigenvalues and eigenvectors, similarity transformation, singular value decomposition, Fourier series, Fourier Transformation, FFT.

Vector Algebra & Calculus: Basic vector algebra; curves; grad, div, curl; line, surface and volume integral, Green's theorem, Stokes's theorem, Gauss-divergence theorem.

Differential Equations: ODE: homogeneous and non-homogeneous equations, Wronskian, Laplace transform, series solutions, Frobenius method, Sturm-Liouville problems, Bessel and Legendre equations, integral transformations; PDE: separation of variables and solution by Fourier Series and Transformations, PDE with variable coefficient.

Numerical Technique: Numerical integration and differentiation; Methods for solution of Initial Value Problems, finite difference methods for ODE and PDE; iterative methods: Jacobi, Gauss-Siedel, and successive over-relaxation.

Complex Number Theory: Analytic function; Cauchy's integral theorem; residue integral method, conformal mapping. Statistical Methods: Descriptive statistics and data analysis, correlation and regression, probability distribution, analysis of variance, testing of hypothesis.

Optimization Techniques: Calculus of several variables, Implicit function theorem, Nature of singular points, Necessary and sufficient conditions for optimization, Constrained Optimization, Lagrange multipliers, Gradient method – steepest descent method.

Texts / Reference Books:

1. H. Kreyszig, "Advanced Engineering Mathematics", Wiley, (2006).
2. Gilbert Strang, "Linear Algebra and Its Applications", 4th edition, Thomson Brooks/Cole, India (2006).
3. J. W. Brown and R. V. Churchill, "Complex Variables and Applications", McGraw-Hill Companies, Inc., New York (2004).
4. J. W. Brown and R. V. Churchill, "Fourier Series and Boundary Value Problems", McGraw-Hill Companies, Inc., New York (2009).
5. G. F. Simmons, "Differential Equations with Applications and Historical Notes", Tata McGraw-Hill Edition, India (2003).
6. S. L. Ross, "Differential Equations" 3rd edition, John Wiley & Sons, Inc., India (2004).
7. K. S. Rao, "Introduction to Partial Differential Equations", PHI Learning Pvt. Ltd (2005).

8. R. Courant and F. John, "Introduction to Calculus and Analysis, Volume I and II", Springer-Verlag, New York, Inc. (1989).
9. K. Atkinson and W. Han, "Elementary Numerical Analysis" 3rd edition, John Wiley & Sons, Inc., India (2004).
10. R. A. Johnson and G. K. Bhattacharya, "Statistics, Principles and Methods", Wiley (2008).

Course Description

Course Objectives

- At the end of this course, students should:
 1. Introduce students to ordinary differential equations and the methods for solving these equations, using differential equations as models for real-world phenomena, and integrate the knowledge accumulated in the calculus to solve applied problems.
 2. To apply the mathematical methods to solve the engineering problem.
 3. Provide a rigorous introduction to upper-level mathematics which is necessary for students of engineering, physical sciences, and mathematics.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-101.1	Identify the powerful mathematical approaches to solve complex engineering problems.
AMTA-101.2	Apply the appropriate mathematical theories and computational tools to solve mathematical problems.
AMTA-101.3	Students can remember terminologies and formulae in matrices, complex numbers, differential calculus, vector Algebra & calculus, differential Equations, Numerical techniques, and complex number theory.
AMTA-101.4	Discuss the appropriate application of mathematical theory in the solution of engineering problems.
AMTA-101.5	Students can find the matrices to solve systems of equations, work with and solve homogeneous and non-homogeneous ODEs and systems of ODEs and recognize ODEs of varying order and use these to solve problems involving population dynamics, oscillation of a spring and resistance in a circuit, etc.

AMTA-102	METAL REMOVAL PROCESSES	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Introduction: Machining Process, Types of machining processes; Chip formation; Orthogonal and Oblique Cutting; Types of Chips; Built-up edge formation. Tool Geometry: Reference planes; Tool specification: American System (ASA), continental or Orthogonal System (ORS), International or Normal Rake system (NRS).

Mechanics of Metal Cutting: Cutting parameters; Mechanism of Orthogonal and Oblique cutting, chipping action Merchant's Circle Diagram; Measurement of cutting Forces, Co-efficient of Friction: Determination of stress, strain and strain rate; Measurement of shear angle; Thin Zone model: Lee and Shaffer's Relationship, processing of polymers, ceramics, and composites.

Mechanisms of wear: Tool Material, Types of tool wear, Abrasion, Adhesion and Diffusion. Progressive tool wear: flank and crater wear.

Tool Life: variables affecting tool life - cutting conditions, tool geometry, Types of tool materials, fabrication of cutting inserts, coatings, work material and cutting fluid, Economic of Machining. Thermal Aspects of Machining.

Introduction to advanced machining processes such as USM, AJM, ECM, EDM, LBM, and EBM. Introduction and differentiation in between the additive and subtractive processes.

Texts / Reference Books:

1. E.J.A. Armarego and R.H.Brown-The machining of Metals
2. G Boothroyd-Fundamentals of Metal Machining and Machine tools
3. A. Ghosh and Asok Mallik- Machining Science
4. G.K. Lal and S.K. Choudhury-Fundamental of Manufacturing Processes
5. M.C. Shaw-Metal Cutting Principle
6. Materials and Processes in Manufacturing, Degarmo, J. T. Black, Prentice Hall of India Pvt Ltd.
7. Manufacturing Processes for Engineering Materials, Kalpakjian and Schmid, Prentice Hall.
8. Principles of Machine Tools, G. C. Sen and A, Bhattacharya, New Central Book Agency
9. Theory of Metal Cutting, A. Bhattacharya.
10. Manufacturing Science: Ghosh and Mallick, East-West Press Private Limited
11. Machining and Metal Working Handbook, Ronal A Walsh and Denis Cormier McGraw Hill Publication.

Course Description

Course Objectives

- At the end of this course, students should:
 1. Apply contemporary knowledge to understand the basic machining processes.
 2. Theoretically and experimentally analyze the force and power required in different machining operations.
 3. Calculate the tool wear and tool life under different cutting environments.
 4. Understand the different types of tool wear mechanisms responsible for tool wear.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-102.1	Understand the different type of machining processes, chip formation, tool geometry and tool specifications.
AMTA-102.2	Analyse the mechanics of metal cutting and calculate the shear stress, strain and strain rate, shear angles.
AMTA-102.3	Calculate the friction applied, cutting forces and cutting temperature in metal cutting processes.
AMTA-102.4	Understand the different tool wear mechanisms involved in machining and estimation of tool life.
AMTA-102.5	Understand the different type of advanced machining processes. Difference between the additive and subtractive processes.

AMTA-103	MATERIALS PROCESSING	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Solidification processing: Plane front solidification, cellular solidification, cellular-dendritic transition, Theories of regular and irregular eutectic growth, Rheocasting, Thixocasting, casting of composites.

Powder processing: preparation of metallic, ceramic and composite powders; Sintering and full density processing; Metal forming processes: Deformation theories, Applications in rolling, forging, extrusion, machining; Processing of new materials: nanomaterials and biomaterial.

Joining of materials: Fundamentals of liquid and solid state joining: friction stir welding, joining of similar and dissimilar materials;

Production of metal powders: recent developments in powder production, mechanical alloying. Development of nanostructures and composite materials via powder processing route. Characteristics of powders. Compaction in rigid dies, hot and cold isostatic compaction.

Processing of ceramics: Crystal Systems, Amorphous Systems - Glass, Phase Equilibrium, Sintering of ceramics, Microstructure of Ceramics, Mechanical Properties, Thermal Properties, Optical Properties, Electrical and Magnetic Properties, Chemical Properties Traditional Ceramic Raw Materials, Non-Traditional and Special Ceramic Raw Materials, Glass ceramics, Bio implants, Advanced ceramics; Material processing from solid waste from metal industry and thermal power plants.

Texts / Reference Books:

1. Porter, Easterling and Sherif, Phase Transformation in metals and alloys
2. Randall M. German, Powder Metallurgy & Particulate Materials Processing
3. WD Kingery, HK Bowen, DR Uhlmann, Introduction to ceramics
4. B.A. Wills and T. Napier-Munn, Wills' Mineral Processing Technology
5. E.G. Kelly and D.J. Spottiswood, Introduction to Mineral Processing

Course Description

Course Objectives

- At the end of this course, students should:
 1. Apply contemporary knowledge to understand the different solidification, joining, and powder processing techniques.
 2. Analyze the different metal forming processes and mineral processing techniques.
 3. To provide the skills to characterize the crystals and amorphous systems and evaluate the different properties.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-103.1	Understand the variety of solidification techniques, different laws and advanced solidifications methods.
AMTA-103.2	Fabrication of different powder processing techniques and study of different applied deformation theories.
AMTA-103.3	Analyze and join the dissimilar structures with advanced joining processes.
AMTA-103.4	Design and development of different mineral processing instruments. Classify, characterize and control the different mineral particles.
AMTA-103.5	Study of different ceramic processing systems. Calculate the mechanical, optical, electrical, chemical and magnetic properties of different ceramic materials.

AMTA-104	SENSORS AND ACTUATORS	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Brief overview of measurement systems, classification, characteristics and calibration of different sensors. Measurement of displacement, position, motion, force, torque, strain gauge.

Pressure flow Temperature sensor sensors, smart sensor. Optical encoder, tactile and proximity, ultrasonic transducers, opto-electrical sensor, gyroscope.

Principles and structures of modern micro sensors, micro-fabrication technologies: bulk micromachining, surface micromachining, LIGA, assembly and packaging.

Pneumatic and hydraulic systems: actuators, definition, example, types, selection. Pneumatic actuator. Electro-pneumatic actuator. Hydraulic actuator, control valves, valve sizing valve selection.

Electrical actuating systems: solid-state switches, solenoids, electric motors, DC motors, Single phase and three-phase induction motor, synchronous motor, stepper motors. Piezoelectric actuator: characterization, operation, and fabrication.

Texts / Reference Books:

1. John G. Webster, Editor-in-chief, "Measurement, Instrumentation, and Sensors Handbook", CRC Press (1999).
2. Jacob Fraden, "Handbook of modern Sensors", AIP Press, Woodbury (1997).
3. Nadim Maluf, "An Introduction to Microelectromechanical Systems Engineering", Artech House Publishers, Boston (2000).
4. Marc Madou, "Fundamentals of Microfabrication", CRC Press, Boca Raton (1997).
5. Gregory Kovacs, "Micro machined Transducers Sourcebook", McGraw-Hill, New York (1998).
6. E. O. Deobelin and D. Manik, "Measurement Systems – Application and Design", Tata McGraw-Hill (2004).
7. D. Patranabis, "Principles of Industrial Instrumentation", Tata McGraw-Hill, eleventh reprint (2004).
8. B. G. Liptak, "Instrument Engineers' Handbook: Process Measurement and Analysis", CRC (2003).

Course Description

Course Objectives

- At the end of this course, students should:
 1. Apply contemporary knowledge to understand the classification of different types of measurement systems their static and dynamic characteristics.
 2. Different types of sensors and actuators with their working principles for different domain of applications.
 3. To provide skills to design a mechatronics system with the integration of sensors and actuators.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-104.1	Understand the variety of measurement systems and their characteristics. Study of some basic sensors.
AMTA-104.2	Study the working principle and development of advanced sensors based on the different applied principles.
AMTA-104.3	Different micro-fabrication technologies for design and development of a sensors and other MEMS devises.
AMTA-104.4	Study of different mechanical actuating system.
AMTA-104.5	Study of different electrical actuating system. Characterization, operation, and fabrication of piezoelectric actuator.

Elective- I

AMTA-111	FRICITION, WEAR AND LUBRICATION	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Introduction: Fundamentals of Tribology, Application of Tribology

Engineering surfaces, nature of surface and surface measurements.

Friction: Origin of Friction; Friction theories; Law of Rolling friction, Junction Growth Theory, Friction Instability.

Wear: Wear Mechanism, Types of wear - adhesive, abrasive, fatigue, Fretting, corrosive etc.; testing methods; Wear analysis; Wear reduction methods.

Lubrication: Basic equations of the theory of lubrication, calculation of oil flow rate; Methods of lubrication; Types of industrial lubricants and their standard grades; Boundary, hydrostatic and hydrodynamic lubrication.

Texts / Reference Books:

1. Principles of Tribology J. Halling
2. Engineering Tribology Prasanth Sahoo
3. Friction & Wear B. Pugh
4. Fundamentals of Tribology Basu, Sengupta, Ahjua
5. Friction, Wear, Lubrication K. C Ludema

Course Description

Course Objectives

- At the end of this course, students should:
 1. Apply contemporary knowledge to understand the fundamentals of tribology involved in the engineering applications
 2. Analyze the wear debris and different wear mechanisms.
 3. To provide skills to calculate the wear and frictional force with wear mechanisms in different fields of engineering applications.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-111.1	Understand the fundamentals of tribology.
AMTA-111.2	Analyse the engineering surfaces, nature of surfaces and its measurements.
AMTA-111.3	Understand the origin of friction, different frictional theories and different measurement methods.
AMTA-111.4	Calculate the wear rate and different types of wear mechanisms between two meeting surfaces.
AMTA-111.5	Study the different types of lubricants and different methods of lubrication. Analyse the effect of lubrication between two meeting surfaces..

AMTA-112	FINITE ELEMENT METHOD	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Introduction to Finite Difference Method and Finite Element Method, Advantages and disadvantages, Mathematical formulation of FEM, Variational and Weighted residual approaches;

Shape functions, Natural co-ordinate system, Element and global stiffness matrix, Boundary conditions, Errors, Convergence and patch test, Higher order elements. Application to plane stress and plane strain problems;

Axi-symmetric and 3D bodies, Plate bending problems with isotropic and anisotropic materials, Structural stability, other applications e.g., Heat conduction and fluid flow problems. Idealisation of stiffness of beam elements in beam-slab problems;

Applications of the method to materially non-linear problems, Organisation of the Finite Element programmes, Data preparation and mesh generation through computer graphics, Numerical techniques, 3D problems.

FEM an essential component of CAD, Use of commercial FEM packages, Finite element solution of existing complete designs, Comparison with conventional analysis.

Texts / Reference Books:

1. Introduction to Finite Elements in Engineering, T. R. Chandrupatla & A. D. Belegundu, 2nd Ed., PHI, 2001.
2. An Introduction to the Finite Element Method, Reddy, J. N., 2005.
3. The Finite Element Method, O. C. Zienkiewicz, 3rd Ed., McGraw-Hill, 1997.
4. The Finite Element Method in Engineering, S. S. Rao, 2nd Ed., Elmsford, Pergamon, 1989.
5. Introduction to the Finite Element Method: A Numerical Method for Engineering Analysis, Desai, C. S. and Abel, J. F., 1972
6. The Finite Element Method O.C. Zienkiewicz and R.L. Taylor McGraw Hill
7. Finite Element Procedure in Engineering Analysis K.J. Bathe McGraw Hill
8. Finite Element Analysis C.S. Krishnamoorthy Tata McGraw Hill
9. Concepts and Application of Finite Element Analysis R.D. Cook, D.S. Malcus and M.E. Plesha John Wiley
10. Finite Element and Approximation O.C. Zenkiewicy & Morgan

Course Description

Course Objectives

At the end of this course, students should:

1. Learn basic principles of finite element analysis procedure.
2. Understand the theory and characteristics of finite elements that represent engineering structures.
3. Apply finite element solutions to structural, thermal, dynamic problems to develop the knowledge and skills needed to effectively evaluate finite element analyses.

Course Outcome

Upon successful completion of this course you should be able to:

AMTA-112.1	Understand the concepts behind formulation methods in FEM.
AMTA-112.2	Develop element characteristic equation and generation of global equation.
AMTA-112.3	Identify the application and characteristics of FEA elements such as bars, beams, plane and iso-parametric elements and able to apply suitable boundary conditions to a global equation for bars, trusses, beams, circular shafts, heat transfer.
AMTA-112.4	Apply boundary equations for nonlinear problems, and data preparation and mesh generation through computer graphics.
AMTA-112.5	Learn different commercial FEM packages and able to apply in solving different complex problems.

AMTA-113	PRECISION ENGINEERING	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Introduction to precision engineering for manufacturing. Role of CAD/CAM in precision manufacturing, and aspects of sustainable manufacturing and design for sustainability.

Top down and bottom up approach, development of Nanotechnology, precision and micro-machining, diamond turning of parts to nanometer accuracy.

Stereo microlithography, machining of micro-sized components, mirror grinding of ceramics, ultra precision block gauges.

Machine tool elements and structure, sources of error, thermal, static, dynamic and process related; precision machining processes and process model.

Diamond turning, fixed and free abrasive processes, sensors for process monitoring and control, metrology, actuators, examples of precision component manufacture.

Texts / Reference Books:

1. Precision Manufacturing, by David Dornfeld and Dae-Eun Lee, Springer, 2008.
2. Dornfeld, D., and Lee, D. E., Precision Manufacturing, 2008, Springer.
- A. H. Slocum, Precision Machine Design, 1992, Prentice-Hall.
3. H. Nakazawa, Principles of Precision Engineering, 1994, Oxford University Press.
4. P. Seyfried, H. Kuntzmann, P. McKeown and M. Weck, eds., Progress in Precision Engineering, Springer-Verlag, 1991.
5. C. Evans, Precision Engineering; an Evolutionary View, Cranfield Press, 1989.
6. Tlusty, J., Manufacturing Processes and Equipment, Prentice-Hall, Upper Saddle River NJ, 2000.
7. Thomas, T. Rough Surfaces, 2nd ed., Imperial College Press, London, 1999.
8. Whitehouse, D. J., Handbook of Surface Metrology, Institute of Physics Publishing, Philadelphia PA, 1994.

Course Description

Course Objectives

At the end of this course, students should:

1. Learn the basics of precision engineering for manufacturing.
2. Analyze how precision engineering is beneficial for utilizing sensors in monitoring and control.
3. Familiarize with skills that need for designing the sustainable product by using precision engineering.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-113.1	Understand the role of precision engineering for manufacturing and learn how precision is important for evaluating the performance of precision machinery for manufacturing.
AMTA-113.2	Learn appropriate technique for controlling and monitoring the source of errors.
AMTA-113.3	Understand different techniques such as diamond turning, fixed and free abrasive processes for manufacturing highly precise component.
AMTA-113.4	Apply the aspects of sustainable manufacturing in designing of highly sustainable model.

AMTA-114	COMPOSITE MATERIALS	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Introduction to composites: Basic concepts, Structural applications, Classification of Organic Matrix composites, Polymer matrix composites (PMC), Carbon matrix Composites or Carbon-Carbon Composites, Metal matrix composites (MMC), Ceramic matrix composites (CMC), Strength and stiffness advantages, manufacturing aspects of composites.

Micro-mechanics: Mechanics of materials method, Bounding methods, Semiempirical methods, Longitudinal and Transverse elastic properties, In plane Shear modulus, Stress – strain relations for General anisotropic, especially orthotropic and transversely isotropic materials, 2D Transformation of Stress – strain, Elastic parameters, Engineering constants, 3D Transformation of Stress – strain, and Elastic parameters.

Macro-mechanics: Strength analysis of Unidirectional Lamina under various loading conditions, Failure theories, Computational procedure for determination of Lamina strength – Tsai – Wu criterion.

Elastic behaviour of Multidirectional laminates – symmetric and asymmetric laminates, Computational procedure for determination of engineering elastic properties, Carpet plots for multidirectional laminates.

Stress and failure analysis of multidirectional laminates, Hygrothermal effects, Experimental methods for characterization and testing of Composite materials. design of laminates.

Texts / Reference Books:

1. Mechanics of composite materials, R. M. Jones, 2nd Edn. Taylor & Francis, 1999.
 2. Engineering mechanics of composite materials, I. M. Daniel & O. Ishai, 2nd Edn., Oxford University Press, 2006.
 3. Principles of composite material mechanics, R. F. Gibson, 2nd Edn. CRC Press, 2007
- MME 52107 Micro-Electro-Mechanical-Systems (MEMS) (3-0-0)
- Introduction to MEMS & materials, fabrication processes, MEMS sensors and actuators, scaling issues in MEMS, electro mechanics, modelling and design techniques, packaging and reliability, measurement techniques for MEMS.

References:

1. MEMS: Introduction and Fundamentals by Mohamed Gad-el-Hak –
2. MEMS mechanical sensors by Stephen Beeby
3. Microsensors, MEMS, and smart devices by Julian W. Gardner, V. K. Varadan, Osama O. Awadelkarim
4. MEMS and microsystems: design, manufacture, and nanoscale engineering- Tai-Ran Hsu –
5. MEMS by N P Mahalik

Course Description

Course Objectives

At the end of this course, students should:

1. Understand the manufacturing processes of reinforcement fibers and matrices for composites.
2. Familiarize with the concept of tailored design methodology.
3. Learn how to estimate the composite stiffness and strength.
4. Understand the objective that provides skills to learn experimental methods for characterization and testing of Composite materials.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-114.1	Identify, describe and evaluate the properties of fibre reinforcements, polymer matrix materials and commercial composites.
AMTA-114.2	Develop competency in one or more common composite manufacturing techniques, and be able to select the appropriate technique for manufacture of fibre-reinforced composite products.
AMTA-114.3	Analyse the elastic properties and simulate the mechanical performance of composite laminates; and understand and predict the failure behaviour of fibre-reinforced composites
AMTA-114.4	Apply knowledge of composite mechanical performance and manufacturing methods to a composites design project.
AMTA-114.5	Critique and synthesise literature and apply the knowledge gained from the course in the design and application of fibre-reinforced composites.

AMTA-115	AUTOMATION AND CONTROL	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Introduction; Mathematical models of physical system; Feedback characteristics of control systems; Control systems and components; Time response analysis, design specifications and performance indices; Concepts of stability and algebraic criteria; Root locus technique; Frequency response analysis; Stability in frequency domain; State variable analysis and design; Controllability and Observability; Optimal control; Adaptive control; Open loop control and closed loop control, Positive and negative feedback, Stepper motor and servo motor, PTP and Continuous control, PID Control Introduction to digital control systems; 8085/8080A Microprocessor architecture and programming.

Texts / Reference Books:

1. Automatic Control Engineering by F.H.Raven, 5th ed., McGrawHill International.
2. Modern Control Engineering by K.Ogata, Prentice Hall.
3. Digital Control Systems by B.C.Kuo, Prentice Hall.
4. Microprocessor architecture, programming, and applications with the 8085/8080A by Ramesh S. Gaonkar, Wiley Eastern Ltd.
5. Fundamentals of Linear State Space Systems by John S. Bay, WCB/McGraw-Hill

Course Description

Course Objectives

At the end of this course, students should:

1. Apply the mathematical theory of control systems to model and control natural engineering systems.
2. Understand the classical and modern control theory and its application.
3. Understand the mathematical relation between a system and its control theory.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-115.1	To create mathematical model of plant or system
AMTA-115.2	To determine stable Operating region of the plant/system
AMTA-115.3	Can design appropriate controller for the plant
AMTA-115.4	Design adaptive controller for plant with variable operating regions
AMTA-115.5	Assess current research issues and future directions in field of the control theory

AMTA-116	THERMO PRODUCTION PROCESSES	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

INTRODUCTION: Need and Classifications of Thermo-Production Processes; **METAL CASTING:** Need and Limitations; Classification of Casting Processes; Sand Mould Casting: Classification of foundry sands; Composition, Properties and Testing of Moulding Sand.

Design of Pattern and Core; Parting Line Design; Gating System Design-Types of gating systems; Design of pouring basin, sprue, runner and in gate; Mould filling velocity and time including friction and velocity distribution in the conduit; Determination of solidification time of castings; Riser design and Placement; Description of Precision Sand Mould Casting Processes; Metal Mould Casting: Preparation of metal mould; Determination of solidification time of castings; Description of Die Casting, Centrifugal casting and Continuous Casting.

METAL WELDING: Need and Limitations; Classification of Welding Processes; Arc Welding-Characteristics of arc and mode of metal transfer; Welding fluxes and coatings; welding machine characteristics-conventional and pulsed power sources, inverter type; Working and Modeling of arc welding processes; Resistance Welding-Principles and technology; power sources; Modeling of resistance welding processes.

Special welding processes; Electron Beam Welding, Laser Beam Welding, Friction Stir Welding, Explosive welding and Ultra-sonic welding, Weldability of cast iron, plain carbon and low alloy steels, stainless steels, Defects and Inspection of welds- NDT; Case studies.

Powder Metallurgy: Introduction, Sintering, Densification and Sizing; Impregnation and Infiltration. Isostatic pressing, Hot pressing and Spark plasma sintering. Porous metals and metal foams, concept of atomisation.

References:

1. Materials and Processes in Manufacturing, Degarmo, J. T. Black, Prentice Hall of India Pvt Ltd.
2. Manufacturing Processes for Engineering Materials, Kalpakjian and Schmid, Prentice Hall.
3. Fundamentals of modern manufacturing processes, M. P. Groover.
4. Manufacturing Science: Ghosh and Mallick, East-West Press Private Limited
5. Machining and Metal Working Handbook, Ronal A Walsh and Denis Cormier McGraw Hill Publication.

Course Description

Course Objectives

- At the end of this course, students should:
 1. Apply contemporary knowledge to understand the classification of thermos production processes involve in manufacturing
 2. Analyze how to design the gating system, pouring basin, sprue, runner and in gate and determine the solidification time of castings.
 3. Understand the different types of welding techniques and powder production methods.
 4. To provide skills to manufacture a part or component with different thermos production processes

Course Outcome

Upon completion of their academic work student will be able:

AMTA-116.1	Understand the classification of thermos production processes. Basics of sand casing and testing of moulding sands.
AMTA-116.2	Design and development of pattern, gating system, pouring basin, sprue and runner. Calculate the solidification time for different types of casting processes.
AMTA-116.3	Understand the conventional welding processes and its process principles.
AMTA-116.4	Study of advanced welding processes. Inspection and identification of welding defects.
AMTA-116.5	Basics about the powder metallurgy. Production, blending, mixing, compacting and sintering of powder.

AMTA-151	LAB-I SENSORS, DRIVES AND CONTROL LAB	L 0	T 0	P 2	C 1
Prerequisite	NIL				

List of Experiments:

1. Introduction about Sensor and its basic working principles.
2. Introduction to basic network theorems & circuits:
 - Norton theorem
 - Thevenin's theorem
 - Wheat stone bridge
 - KLC, KVL, and Maximum Power Transfer Theorem.
3. Introduction of Position, Proximity, Distance & Velocity Measuring Sensors. (Ultrasonic Sensor, IR Sensor, Radar Sensor, LVDT etc.)
4. Introduction of Different Touch and Pressure Type Sensor.
5. Introduction of Force and Torque Measuring Sensor.
6. Introduction of Temperature Sensor. (To Measure the Characteristics of RTD, Thermocouple and LM35 Sensor)
7. Introduction of Flow Measuring Sensor.
8. Introduction of different rays, Waves and chemical Detecting Sensors. (Color, LDR, solar, etc.)
9. Introduction to different Actuators (Electrical, Pneumatic, Hydraulic). (SMA, DEA, Piezoelectric, motors)
10. DIY (Do It Yourself) project on sensors and actuators model/prototype design/development; using these develop a system contain least two sensors and two actuators.

Text Books:

- [1]. John G. Webster, Editor-in-chief, "Measurement, Instrumentation, and Sensors Handbook", CRC Press (1999).
- [2]. Jacob Fraden, "Handbook of modern Sensors", AIP Press, Woodbury (1997).
- [3]. Nadim Maluf, "An Introduction to Microelectromechanical Systems Engineering", Artech House Publishers, Boston (2000).

Course Objectives

At the end of this course, students should:

1. Apply contemporary knowledge for understanding the operation of various sensors used in automation industry.
2. Develop the ability to analyze, and debug the electrical circuits using fundamental circuit theory.
3. Develop the skill to design circuits for interfacing the sensors to solve real world problems.

Course Outcome:

Upon completion of their academic work student will be able:

AMTA -151.1	To understand the basic working principle of sensors and drives systems.
AMTA -151.2	To understand the basic network theorems and analyse the drive and sensor interfacing circuits.
AMTA -151.3	To understand the working principle of different touch and pressure type sensors, and the design of sensor interfacing circuits.
AMTA -151.4	Analyse how Electrical, Pneumatic, and Hydraulic actuators operate and develop the ability to debug various sensors and actuator circuits.
AMTA -151.5	Develop the prototype of real world automation project using sensors and actuators.

AMTA -152	MANUFACTURING LAB	L 0	T 0	P 2	C 1
Prerequisite	NIL				

List of Experiments:

1. To study the basics of turning, milling, drilling and grinding operations.
2. To study the different operations performed on lathe machine, including working principle of cutting force measurement with the help of the dynamometer.
3. Measurement and analysis of cutting forces in orthogonal turning operation.
4. To measure the machining temperature during orthogonal cutting.
5. To study the effects of grinding variables on grinding forces, specific energy and surface finish.
6. To study the various features of Micro-Machining Machine and study the different parameters of machine.
7. Writing a part program for different turning on CNC lathe.
8. Writing a part program for profile cutting operation on CNC milling machine.
9. To study the various features of SJ-301 surface roughness measuring machine. Also measure the surface roughness parameters of the specimen.
10. To study the effects of EDM variables on MRR, surface finish and TWR.
11. To study the effects of FDM 3D printing parameters.
12. To study the electric induction furnace for melting of different materials.

Course Objectives

At the end of this course, students should:

1. Understand practical orientation of manufacturing processes.
2. Knowledge on different kinds of production processes and practices available for shaping several daily used parts for industries.
3. Prepare assembly drawings, sectional views and bill of materials for selection of equipment's for various manufacturing processes will be understood.

Course Outcome:

Upon completion of their academic work student will be able:

AMTA -152.1	Students able to learn the different machining operations on lathe and other machines such as milling and drilling and grinding.
AMTA -152.2	Measure and determine the cutting forces and tool temperatures in different machining operations.
AMTA -152.3	Students able to familiarize with techniques that used to evaluate the metal removal rate (MRR) and tool wear rate (TWR) in electric discharge machining.
AMTA -152.4	Students able to recognize the different 3D printing techniques and study the principle of working of fused deposition modelling (FDM).
AMTA -152.5	Students understand to measure the surface roughness on work piece during different machining processes.

AMTA-153	INDEPENDENT STUDY AND RESEARCH-I (ISR-I)	L 0	T 0	P 2	C 1
Prerequisite	NIL				

Course Objectives

At the end of this course, students should:

1. The student will learn to work on research problems.
2. The basic purpose of this lab is to prepare students for good research works.
3. One of the objectives is also to help students in selection of their thesis dissertation topic.

Course description:

1. Student will learn how to read technical papers and how to analyse them.
2. Students will identify a small problem to work with after literature survey.
3. Students will prepare a detailed report of the works done by him.
4. Progress will be checked once a month for support and direction. A mentor will be assigned to each student for supervision.

Course Outcome:

Upon completion of their academic work student will be able:

AMTA-153.1	Understand basic concepts of research.
AMTA-153.2	Apply previous literature review in selection of a workable research problem.
AMTA-153.3	Analyze the methods through which a research problem can be solved.
AMTA-153.4	Design a scheme to do work on research problem and to achieve targets.
AMTA-153.5	Assess challenges in doing research work based on literature reviews. .

SEMESTER-II
CORE COURSES

AMTA-201	FUNDAMENTAL OF ROBOTICS & AUTOMATION	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Introduction: Automation in Production System, Principles and Strategies of Automation, Basic Elements of an Automated System, Advanced Automation Functions, Levels of Automations. Production Economics. Automated Flow lines, Methods of Work Part Transport, Transfer Mechanism, and Buffer Storage.

Material handling and Identification Technologies: Types of Material Handling Equipment, Conveyor Systems, Automated Guided Vehicle Systems. Automated Storage Systems: Storage System Performance, Automated Storage/Retrieval Systems, Product identification system: Barcode, RFID etc.

Automated Assembly Systems: Types of Automated Assembly Systems. Different control technologies in automation. Automated Inspection and Testing, Automated Inspection Principles and Methods, Sensor Technologies for Automated Inspection, Coordinate Measuring Machines, Machine Vision, and Other Optical Inspection Methods.

Fundamentals of robotics. Introduction, construction and applications. The robot and its peripherals, Control systems and components, robot motion analysis and control, end effectors, feedback systems. Jacobian, work envelopes. Trajectory planning, Joint Interpolated Trajectory, Link joints and their Manipulator dynamics and force control.

Sensors in Robotics: Developments in sensor technology, sensory control Vision, ranging, laser, acoustic, tactile.

Programming Language: Different programming languages. Mobile robots, walking devices. Robot reasoning.

Robot applications: Application of robots in surgery, Manufacturing industries, space and underwater. Humanoid robots, Micro robots, Social issues and Future of robotics.

Texts / Reference Books:

1. "Automation, Production Systems and Computer Integrated Manufacturing"- M. P. Grover, Pearson Education.
2. "Computer Based Industrial Control" – Krishna Kant, EEE-PHI
3. Principles and Applications of PLC – Webb John, Mcmillan 1992
4. "An Introduction to Automated Process Planning Systems" – Tiess Chiu Chang & Richard A. Wysk
5. "Anatomy of Automation" – Amber G.H & P.S. Amber, Prentice Hall.
6. K.S Fu, R.C. Gonzalez, C.S.G. Lee, Robotics, McGraw Hill, 1987.
7. Y. Koren, Robotics for Engineers, McGraw Hill, 1985.
8. J.J. Craig, Robotics, Addison-Wesley, 1986.
9. Saeed B. Niku, "Introduction to Robotics – Analysis, Systems and Application": PHI 2006.
10. Richard D, Klafter, Thomason A Chmiel Owski, Michel Nagin "Robotics Engg-an Integrated Approach" PHI 2005.

11. R.K. Mittal & I.J. Nagrath, “Robotics & Control” TMH-2007.
12. Groover. M.P. Industrial Robotics, technology, programming and application Mc-Graw Hill 2012.
13. S. K. Saha, “Introduction to Robotics”, Tata McGraw-Hill Publishing Company Ltd. (2008).

Course Description

Course Objectives

- At the end of this course, students should:
 1. To provide an introduction to Robotics and Automation including robot classification, design and selection, analysis and applications in industry.
 2. To provide information on various types of end effectors, their design, interfacing and selection.
 3. To provide the details of operations for a variety of sensory devices that are used on robot, the meaning of sensing, classification of sensor, that measure position, velocity & acceleration of robot joint.
 - 4 To familiarize the basic concepts of transformations performed by robot.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-201.1	Understand basic components of robotics, classification of robots and their industrial applications
AMTA-201.2	Know on types of robot grippers, their usage and design considerations.
AMTA-201.3	Understand about various types of sensory devices their working principle and applications.
AMTA-201.4	Apply basic homogeneous transformations related to the movement of manipulator end effector.
AMTA-201.5	Design a robot mechanism to meet kinematics requirements and to write simple programs to control it.

AMTA-202	MICRO MANUFACTURING PROCESSES	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Introduction to Advanced Manufacturing Processes: Overview of non-conventional machining processes with (AJM, USM, ECM, EDM, EBM, LBM, AFM, MRF, MAF, MFP and MRAFF etc.)

Introduction to use of non-conventional processes for micro-machining. Introduction to nano finishing and need of nano finishing.

Introduction to Water jet machining (WJM) and Water Abrasive jet machining (WAJM): Basic principle and MRR estimation.

Abrasive Jet Machining (AJM): Introduction to abrasive jet machining (AJM), Mechanics of AJM, AJM process parameters Components of AJM (Abrasive, Gas, Setup), Mixing and Mass ratio and Material Removal Rate (MRR). Water Abrasive jet machining (WAJM): Introduction to WAJM, Basic principle and MRR estimation.

Ultrasonic Machining (USM): Basics of USM processes, Mechanics of USM, Process parameters of USM, Dependence of process parameters in estimation of MRR, Ultrasonic machining setup, Design of acoustic ultrasonic head and feed mechanism in USM.

Basic Principle of Electrochemical Machining, Estimation of MRR in ECM, MRR in multiphase alloys, Surface Finish in ECM of alloys, Basics of Electrochemical drilling, Basics of Electrochemical Grinding, Electro stream drilling and Electrochemical Grinding, Allied Processes, Electrochemical turning, Electrochemical Milling, Electrochemical deburring, Electrochemical boring etc.

Electro-discharge machining (EDM), Process parameters of EDM, Mechanics of EDM, Theoretical estimation of MRR in EDM, Role of cavitation in material removal in EDM, Role of melting temperature of the work-piece material, EDM circuits and operating principles, Electrode and dielectric fluid, EDM turning, Wire EDM. Comparison of EBM machining with other thermal processes, Setup for EBM, Mechanics of EBM process, Types of lasers and feedback mechanisms in Lasers, Mechanics of material removal in Laser machining.

Texts / Reference Books:

1. Advanced manufacturing processes, Hassan Abdel, Gabad El Hoffy, McGraw Hill.
2. V.K. Jain, Advance Machining Processes, Allied Publisher Bombay.
3. Ghosh and Mallik, Manufacturing Science, EWP Private Ltd.
4. Pandey P.C., Shan H.S., Modern machining processes, Tata McGraw-Hill Education.
5. Weller E.J., Non-traditional machining processes, Society of Manufacturing Engineers, Publications.
6. The Science and Engineering of Micro-fabrication, Stephen P. Campbell, Oxford University press.

Course Description

Course Objectives

- At the end of this course, students should:
 1. Apply contemporary knowledge to understand the different unconventional manufacturing processes
 2. Analyze how to select the available unconventional manufacturing processes for Variety of engineering applications
 3. To provide skills to machine the components and evaluate the machining performance.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-202.1	Understand the classification of unconventional machining processes. Basics of nano finishing processes.
AMTA-202.2	Working principle and calculation of MMR in abrasive jet machining and water abrasive jet machining processes.
AMTA-202.3	Working principle and calculation of MMR in ultrasonic machining process. Design of different parts of ultrasonic machining process.
AMTA-202.4	Working principle and calculation of MMR in electrochemical machining process and basics of its allied processes.
AMTA-202.5	Working principle and calculation of MMR in electric-discharge machining, electron beam machining and laser beam machining processes.

AMTA-203	ADVANCED CHARACTERIZATION TECHNIQUES	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Definition of Microscopy and spectroscopy and difference, Electromagnetic spectrum and range of different rays, Principles, instrumentation, design, analysis and application of UV-visible and IR spectroscopy, mass spectrometry, Mossbauer and NMR spectroscopy.

Basics of Diffraction, X-ray methods of analysis including powder diffraction, wavelength and energy dispersive x-ray fluorescence, instrumentation and application

Optical microscopy, Electron-sample interaction, Electron microscopy: basics, analysis and application, microprobe, scanning electron microscopy (SEM), ESCA and Auger techniques: basics and analysis, photo electron spectroscopic methods: XPS and UVS, basics and analysis, scanning tunnelling microscopy: concepts and application, atomic force microscopy: basics, modes and application.

Chromatography and their types, Column chromatography, HPLC, Thin layer chromatography, thermal analysis including DTA, DSC and TGA.

Hands on training on preparation of fine particles, single crystals, growth of single crystals and thin films and their thermal analysis, magnetic measurements, X-ray diffraction, Transmission electron microscopy (TEM) and analyses, electrical and dielectric measurements.

Texts / Reference Books:

1. George M. Crankovic (Ed.), ASM Handbook: Volume 10: Materials Characterization, ASM International, 1986
2. Robert Cahn (Ed.), Concise Encyclopaedia of Materials Characterization, 2nd Edition, Elsevier, 2004
3. Charles Evans, Richard Brundle and Shaun Wilson, Encyclopaedia of Materials Characterization: Surfaces, Interfaces, Thin Films, Elsevier, 1992
4. Tantra, Ratna, Nanomaterial Characterization: An Introduction, Wiley (2016)

Course Description

Course Objectives

- At the end of this course, students should:
 1. Apply contemporary knowledge to understand the different characterization techniques
 2. Analyze the data recorded with the advanced characterization techniques
 3. To provide skills to select a suitable characterization technique for the variety of engineering applications

Course Outcome

Upon completion of their academic work student will be able:

AMTA-203.1	Understand the design, principle and application of different characterization instruments such as UV, visible and IR spectroscopy, mass spectrometry, Mossbauer and NMR spectroscopy.
AMTA-203.2	Analysis of powder diffraction, wavelength and energy dispersive using x-ray methods.
AMTA-203.3	Study the working principle and analysis using electron microscopy, scanning tunneling and atomic force microscopy.
AMTA-203.4	Study and working principle of chromatography, thermal analysis including DTA, DSC and TGA.
AMTA-203.5	Hands on training on preparation of fine particles and its analysis using X-ray diffraction, SEM and TEM analyses etc.

ELECTIVE- (II-II1)

AMTA-211	ADDITIVE MANUFACTURING	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Introduction: Historical developments, Fundamentals of RP/AM Systems and its Classification, Rapid prototyping process chains, 3D modelling and mesh generation, Data conversion and transmission.

RP Database: Rapid prototyping data formats, STL format, STL file problems, STL file repair, Network based operations, Digital inspection, Data warehousing and learning from process data; Computer aided design (CAD) and manufacturing (CAM); AM Different AM processes and relevant process physics AM process chain.

Different Rapid Prototyping Processes – namely Stereolithography (STL), Laser Additive Manufacturing.; Fused Deposition Modelling (FDM), Solid Ground Curing (SGC); Laminated Object Manufacturing (LOM) Ballistic Particle Manufacturing (BPM), Three Dimensional Printing (3DP), numerical Examples; Electron Beam based Additive Manufacturing; strategies and post processing. Monitoring and control of defects.

Introduction to reverse engineering Traditional manufacturing via AM, Direct processes – Rapid Prototyping, Rapid Tooling. Rapid Manufacturing; Indirect Processes - Indirect Prototyping. Indirect Tooling, Indirect Manufacturing. Comparison between RP processes and CNC Machining.

RP interfacing with Manufacturing processes - Investment Casting, Vacuum Casting, and Rapid manufacturing processes like - Vacuum Casting Processes via RP Processes, Electroplated Prototypes, RTV Silicon Moulds, Direct Inkjet SL Tooling.

Texts / Reference Books:

1. Rapid Manufacturing: An Industrial Revolution for the Digital Age. Neil Hopkinson, Richard Hague, Philip Dickens (Editors); Wiley; Jan., 2006; ISBN: 10: 0470016132; 13: 978-0470016138.
2. Additive Manufacturing Technologies; Rapid Prototyping to Direct Digital Manufacturing. Ian Gibson, David W. Rosen, Brent Stucker; Springer; January, 2010; ISBN: 978-1-4419-1119-3.
3. Rapid Prototyping: Principles and Applications. Rafiq I. Noorani; Wiley; Oct., 2005; ISBN: 10: 0471730017; 13: 978-0471730019.
4. User's Guide to Rapid Prototyping. Todd Grimm; Society of Manufacturing Engineers; February, 2004; ISBN: 0-87263-697-6.
5. Rapid Prototyping - Laser-based and Other Technologies. Patri K. Venuvinod and Weiyin Ma; Kluwer Academic Publishers; October, 2003; ISBN: 1-4020-7577-4.

6. Rapid Prototyping. Andreas Gebhardt; Hanser Gardner Publications; 1st ed., June 1, 2003; (Originally published in German, 1995); ISBN: 156990281X.
7. Rapid Prototyping: Principles and Applications. Chua Chee Kai, Leong Kah Fai, Lim Chu-Sing; World Scientific Pub Co; March, 2003; ISBN: 9812381171.
8. Rapid Prototyping: Theory and Practice. Ali Kamrani, Emad Abouel Nasr (Editors); Springer; 1st ed., Jan., 2006; ISBN:10: 0387232907; 13: 978-0387232904.
9. Rapid Manufacturing: The Technologies and Applications of Rapid Prototyping and Rapid Tooling, D.T. Pham, S.S. Dimov.
10. Understanding additive manufacturing: rapid prototyping, rapid tooling, rapid manufacturing By Andreas Gebhardt. Hanser Publishers, 2011.
11. Rapid Prototyping, Tooling and Manufacturing By R. J. M. Hague, P. E. Reeves, Paperback, 2002.
12. Rapid Prototyping Technology: Selection and Application By Kenneth Cooper, CRC, 2001.
13. Rapid Prototyping: Theory and Practice By Kamrani A., Nasr E. A., Springer, 2006
14. Laser assisted fabrication of materials By J.D. Majumdar and I. Manna. Springer Series in Material Science, 2013.
15. Rapid Prototyping: Laser-Based and Other Technologies By Patri K. Venuvinod, Weiyin Ma, Springer, 2004.
16. Rapid Prototyping By Andreas Gebhardt, Hanser, 1996.
17. Rapid Prototyping and Engineering Applications: A Toolbox for Prototype ... By Frank W. Liou, CRC Press, 2007.

Course Description

Course Objectives

At the end of this course, students should:

1. Learn the basics of additive manufacturing/rapid prototyping and its applications in various fields, reverse engineering techniques.
2. Train with different processes in rapid prototyping systems.
3. Understand about mechanical properties and geometric issues relating to specific rapid prototyping applications.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-211.1	Demonstrate the knowledge and basics of Additive Manufacturing and Rapid Prototyping technologies
AMTA-211.2	To understand the different types of error formed in STL file formation and how to overcome this error.
AMTA-211.3	Describe different RP techniques and its manufacturing defects, and control.
AMTA-211.4	To learn about the different range of materials used in additive manufacturing.
AMTA-211.5	This course makes students capable to manufacture any complex parts using additive manufacturing technique and provides understanding regarding the use of different RP interface system.

AMTA-212	THEORY OF METAL FORMING	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Introduction: Review of two dimensional stress and strain, state of stress in three dimensions, Stress tensor, Invariants, Mohr's circle for 3-dimensional state of stress, strain at a point Mohr's circle for strain, Hydrostatic & Deviator components of stress, Elastic stress strain relations, Plastic Anisotropy and Visco plasticity; Concept of solid and flow formulations.

Conventional Forming Processes: Plastic Deformation Analysis Techniques-Slab Method, Upper Bound Method and Slip Line Method; Analysis (using Slab, Upper Bound and Slip Line Method) of: Plain strain Problems: Drawing of Sheet and Extrusion of Plate, Rolling of Plate and Forging of Strip; Axisymmetric Problems: Drawing of Wire and Extrusion of bar and tube, Forging of Solid and Hollow Disc; Bending and Deep Drawing;

Unconventional Forming Processes: Classification; Process Principle, Applications, Equipment, Process Analysis and Die Design of Explosive Forming; Electro-Magnetic Forming; Electro-Hydraulic Forming; Laser Beam Bending and Laser Assisted Deep Drawing;

Micro Forming Processes: Classification; Process Principle and Applications of Conventional Micro-Forming Processes and Unconventional Micro-Forming Processes.

Texts / Reference Books:

1. Materials and Processes in Manufacturing, Degarmo, J. T. Black, Prentice Hall of India Pvt Ltd.
2. Manufacturing Processes for Engineering Materials, Kalpakjian and Schmid, Prentice Hall.
3. Fundamentals of modern manufacturing processes, M. P. Groover.
4. Manufacturing Science: Ghosh and Mallick, East-West Press Private Limited
5. Machining and Metal Working Handbook, Ronal A Walsh and Denis Cormier McGraw Hill Publication.

Course Description

Course Objectives

At the end of this course, students should:

1. Familiarize with various metal forming processes.
2. Learn the role of plastic deformation during forming processes.
3. Understand the different laws and equations used for solving metal forming problems.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-212.1	Recognize the various metal forming techniques
AMTA-212.2	Learn the theory of plasticity and its application for analyzing various metal forming processes.
AMTA-212.3	Analyse the axisymmetric problems such as Drawing of Wire and Extrusion of bar and learn to solve the problems associated with in plastic deformations.
AMTA-212.4	Familiar with different unconventional metal forming techniques.
AMTA-212.5	Understand the role of micro forming processes.

AMTA-213	MACHINE TOOL CONTROL AND CONDITION MONITORING	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Overview of automatic control in machine tools: Open loop and closed loop systems, process model formulation, transfer function, block diagram representation of mechanical, pneumatic and electrical systems, mathematical modelling of mechanical, pneumatic and electrical systems, control actions.

Drive systems and feedback devices in machine tools: Hydraulic and Pneumatic drives, Electrical drives, A.C. Motor, D.C. Motor, Servo motor and Stepper motor. Feedback devices- Syncro, resolver, diffraction gratings, potentiometer, applications of drives in machine tools.

Adaptive control and PLC: Types– ACC, ACO, Applications- adaptive control for turning, milling, grinding and EDM, Programmable logic Controller-Functions-Applications in machine tools. PLC ladder logic diagram, Real time problems using PLC ladder logic, Traffic light control using PLC.

Vibration, acoustic emission / sound: Primary & Secondary signals, Online and Off-line monitoring. Fundamentals of Vibration, Sound, Acoustic Emission. Machine Tool Condition Monitoring through Vibration, Sound, Acoustic Emission, Case Studies.

Condition monitoring, through other techniques: Visual & temperature monitoring, Leakage monitoring, Lubricant monitoring, condition monitoring of Lube and Hydraulic systems, Thickness monitoring, Image processing techniques in condition monitoring.

Texts / Reference Books:

1. MikellP.Groover, “Automation Production system and Computer Integrated Manufacturing”, Prentice Hall of India Pvt. Ltd., 2010
2. Sushil Kumar Srivstava, “Industrial Maintenance Management” S.Chand & Company Ltd., New Delhi, 1998.
3. Manfred Weck, “Hand Book of Machine Tools” – Vol.3, John Wiley & Sons, 1995.

Course Description

Course Objectives

- At the end of this course, students should:
 1. Control and monitoring of machine tools used in production system
 2. To impart basic knowledge and importance on Vibration Based Condition Monitoring in Engineering Fields among the students.
 3. To create the awareness on Vibration Based Condition Monitoring in Research and Application area

Course Outcome

Upon completion of their academic work student will be able:

AMTA-213.1	Understand open loop and closed loop system in machine tools- process.
AMTA-213.2	Apply the knowledge of sensors in Feedback devices such as Syncro, resolver, diffraction gratings, potentiometer, Inductosyn and encoders-application in machine tools
AMTA-213.3	Knowledge will be gained on application and utility of Vibration Based Condition Monitoring used in various sectors and fields
AMTA-213.4	Can perform Machine Tool Condition Monitoring through Vibration, and Sound
AMTA-213.5	Can perform Condition monitoring of machine tools through other techniques such as visual, temperature, leakage, Lubricant etc.

AMTA-214	LASER PROCESSING OF MATERIALS	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Main industrial lasers: He-Ne, CO₂, Excimer, Nd:YAG, Diode, Fiber and Ultra-short pulse lasers and their output beam characteristics; laser beam delivery systems. Overview of Industrial & Scientific Applications of laser: Metrological applications, Holography, Laser Isotope Separation, Laser fusion. Laser processing fundamentals: Laser beam interaction with metal, semiconductor and insulator, Ultra-short laser pulse interaction, heat flow theory and metallurgical considerations. Laser Material Processing Applications: Laser cutting and drilling: Process characteristics, material removal modes, practical performances. Laser welding: Process mechanisms like keyhole and plasma effect, operating characteristics and process variation. Laser surface modifications: Heat treatment, surface re-melting, surface alloying and cladding, surface texturing, LCVD and LPVD. Laser rapid manufacturing. Laser metal forming: Mechanisms involved including thermal temperature gradient, buckling, upsetting. Laser peening: Fundamentals of Laser Shock Processing, Effects of various laser and process parameters, Mechanical effects and microstructure modification during laser shock processing. Theoretical modelling of laser material processing. On-line Process monitoring & control: Laser and process parameters, and work piece characteristics. Economics of Laser Applications in Manufacturing. Laser Safety: Laser safety standards and safety procedures.

Texts / Reference Books:

1. Peter Schaaf, Laser Processing of Materials Book Subtitle, Publisher Springer-Verlag Berlin Heidelberg, 2010.
2. John Ion. Laser Processing of Engineering Materials, 1st Edition, Principles, Procedure and Industrial Application, 2005.
3. Grigorants. Basics of Laser Material Processing. 1st Edition, CRC Press, 1994.
4. W. W. Duley, Laser Processing and Analysis of Materials, Springer.
5. Jacques Perriere, Eric Millon, Eric Fogarassy. Recent Advances in Laser Processing of Materials. 2006.
6. Leonard R. Migliore. Laser Materials Processing, 1st Edition, 1996.
7. Elijah Kannatey-Asibu. Principles of Laser Materials Processing. John Wiley and Sons Ltd, 2009.
8. John Dowden. The Theory of Laser Materials Processing: Heat and Mass Transfer in Modern Technology. Edition 1, Springer 2009.

Course Description

Course Objectives

- At the end of this course, students should:
 1. Introduce the participants to fundamental properties of laser beams as advanced materials processing and manufacturing tool
 2. Build broad understanding of laser based physical processes and their implications in material processing and manufacturing processes
 3. Provide exposure to practical problems and their solutions through laser based manufacturing processes in various industries

Course Outcome

Upon completion of their academic work student will be able:

AMTA-214.1	Understand Laser Operation Mechanism, Properties of Laser Radiation.
AMTA-214.2	Develop understanding on Laser Materials Interactions: Absorption of Laser Radiation, Thermal Effects
AMTA-214.3	Apply Laser based Manufacturing methods such as Laser Casting, Laser Forming/Shaping, Laser Joining for manufacturing of various products
AMTA-214.4	Design and optimize laser based manufacturing process for potential application
AMTA-214.5	Innovative Applications of Laser Materials Processes in Manufacturing

AMTA-215	BIOMATERIALS PROCESSING AND APPLICATIONS	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Introduction: Classes of materials used in medicine: metals, polymers, FRPs, fabrics, nanocomposites, bio-resorbable and bioerodable materials, ceramics, glasses;

Mechanical Properties of Biomaterials; Metals, polymers, and ceramic; acute and chronic responses to implanted biomaterials, drug delivery and tissue engineering; Property requirement of biomaterials;

Concept of biocompatibility; Cell-material interactions and foreign body response; Assessment of biocompatibility of biomaterials, important bio-metallic alloys;

Ti-based, stainless steels, Co-Cr-Mo alloys; Processing and properties of different bio-ceramic materials with emphasize on hydroxyapatite;

Synthesis of biocompatible coatings on structural implant materials; Surface chemistry and physics of selected metals, polymers, and ceramics, surface characterization methodology, surfaces, biosensors and microarrays. Microstructure and properties of glass ceramics; Biodegradable polymers; Design concept of developing new materials for bio-implant applications.

Texts / Reference Books:

1. TeohSwee Hin Engineering Materials for Biomedical Applications (Biomaterials Engineering and Processing Series
2. Ochsner Andreas. Biomaterials and Their Applications, Springer.
3. Qi-Zhi Chen, George Thouas. Biomaterials: A Basic Introduction. CRC Press; 1 edition, 2018.
4. Paul K. Chu, Xuanyong Liu. Biomaterials Fabrication and Processing Handbook. CRC Press Inc,

Course Description

Course Objectives

At the end of this course, students should:

1. Understand common use of biomaterials as metals, ceramics and polymers and its chemical structure, properties, and morphology
2. Understand and account for methods for categorization of biomaterials.
3. Explain methods to modify surfaces of biomaterials and choose material for desired biological response.
4. Describe interactions between biomaterials, proteins and cells.
5. Understand the interaction between biomaterial and tissue for short term and long term implantations, distinguish between reactions in blood and in tissue.
6. Explain the types of material used to replace different organs & tissues of human body.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-215.1	Appreciate how the varying characteristics of the four types of biomaterials (metallic, polymeric, ceramic and composite) affect their properties and interactions with the body.
AMTA-215.2	Apply knowledge of the basic features of bio implant, cell and biomolecule structure and function to evaluate how biomaterial properties influence their interaction with the body.
AMTA-215.3	Discuss the potential applications of biomaterials in combating the most threatening human diseases and appreciate the biomaterial selection requirements.
AMTA-215.4	Understand the potential applications of different bio-ceramic materials with emphasize on hydroxyapatite.
AMTA-215.5	Research the latest developments in biocompatible coatings on structural implant materials related application and design concept of developing new materials for bio-implant applications.

AMTA-216	SURFACE ENGINEERING	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Surface engineering: definition, concept and Importance, Classification of surface modification techniques, Advantages and their Limitations. Surface degradation: basics, causes, types and consequences of surface degradation, Forms of wear – Adhesive, Abrasive, Surface fatigue, Corrosive, Fretting and erosive wear, Classical governing laws related to wear, Techniques to evaluate the wear damage. Corrosion: basics and their types, Corrosion control methods, Corrosion measurements.

Materials characteristics, their importance in surface engineering, wear resistant materials, Selection of materials for engineering the surfaces for specific applications, some case studies; Coatings, new coating concepts including multi-layer structures, Functionally gradient materials (FGMs): basics, types and application, Intermetallic barrier coatings and Thermal barrier coating: basics, types and application.

Principles and application of weld surfacing: SMAW, SAW, GMAW, Thermal spraying: concepts, application and types – Flame spraying, Electric arc spraying, Plasma spraying, Detonation gun spraying and High velocity oxy fuel spraying.

Ion implantation: concepts and application, Chemical Vapour Deposition (CVD): basics, analysis and application, Physical Vapour Deposition (PVD): basics, types, analysis and application, Hardening, conventional Hardening methods like Carburizing, Nitriding, Cyaniding, Advanced hardening methods like Plasma nitriding and Plasma cyaniding, Lasers: types and application, Laser cladding, Alloying, Glazing, Laser and Induction hardening, Heat treatment of steel and re-melting by Laser / TIG.

Microwave: basics and application, Microwave heating, Hybrid Microwave heating, Microwave glazing; Characterization, Importance of Different characterisation techniques – Physical, Mechanical and Functional characterisations, Surface roughness and Surface finish, Micro hardness, Nano Hardness, Strength and Tribological characterizations; Electro deposition and Electroless coatings; and Pulsed Laser Deposition: definition and basics.

Texts / Reference Books:

1. Surface Engineering of Metals: Principles, Equipment, Technologies by T. Burakowski and Wierzchoń T., CRC Press, Boca Raton, Florida, 1999.
2. Surface Engineering Casebook by J.S. Burnell-Gray and P.K. Datta (eds.), Woodhead Publishing Limited, Cambridge, England, 1996.
3. Engineering coatings - design and application by S. Grainger and J. Blunt (eds.), Abington Publishing, Cambridge, England, 1998.
4. Advanced Surface Coatings: A Handbook of Surface Engineering by D. S. Rickerby and A. Matthews (eds), Blackie, London, 1991.
5. Coatings Tribology: Properties, Techniques and Applications in Surface Engineering by K. Holmberg and A. Matthews, Elsevier Science B.V., Amsterdam, 1994.

Course Description

Course Objectives

- At the end of this course, students should:
 1. Apply contemporary knowledge to understand the different surface modification techniques
 2. Analyze the materials surface characteristics for coatings, machining, functionally graded materials and intermetallic coatings etc.
 3. To provide skills to select the suitable engineering material for the variety of engineering applications

Course Outcome

Upon completion of their academic work student will be able:

AMTA-216.1	Understand classification of surface modification techniques. Analysis and evaluation of different types of wear.
AMTA-216.2	Selection of materials for different engineering application. Development of coatings and functionally gradient materials.
AMTA-216.3	Study the working principle and application of different welding and surface modification processes.
AMTA-216.4	Understanding the working principle of advanced thin film deposition techniques and surface enhancement with thermal based processes.
AMTA-216.5	Analysis of microwave glazing techniques and its physical, mechanical and functional characterization.

AMTA-217	MODELLING AND SIMULATION OF MATERIALS	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Basics of modelling and simulations, Empirical and phenomenological modelling, Population balance models, Kinetic models, Stochastic models, Matrix models, Discrete size discrete time models, Discrete size continuous time models.

Continuous size continuous time models, Modelling of flotation networks and simulation of complex flotation circuits, Material balance over complex minerals flow sheets, Physical modelling,

Mathematical modelling, Data modelling as a new type of modelling, Reverse Monte-Carlo Analysis (RMCA): Reconstruction of 3-D atomic ensemble from diffraction data, Discrete Element Modelling (DEM),

Finite Element Method (FEM) and its application in materials science, Ab-initio simulations, Phase-field modelling.

Texts / Reference Books:

1. K. Janssens, G. Frans, D. Rabbe, B. Nestler, E. Kozeschnik, M. Miodownik, Computational Materials Engineering- An Introduction to Microstructure Evolution.
2. K. KesavaRao, Prabhu R. Nott, An Introduction to Granular Flow.
3. Dominik Marx, JürgHutter, Ab Initio Molecular Dynamics: Basic Theory and Advanced Methods.

Course Description

Course Objectives

At the end of this course, students should:

1. Define the basics of simulation modeling and replicating the practical situations in organizations.
2. Develop simulation model using monte-carlo analysis (RMCA) and discrete Element Modelling (DEM).
3. Familiar with the concepts of finite element method (FEM) and its application in materials science

Course Outcome

Upon completion of their academic work student will be able:

AMTA-217.1	Describe the role of important elements of discrete event simulation and modeling paradigm.
AMTA-217.2	Conceptualize real world situations related to systems development decisions, originating from source requirements and goals.
AMTA-217.3	Develop skills to apply simulation software to construct and execute goal-driven system models.
AMTA-217.4	Interpret the model and apply the results to resolve critical issues in a real world environment.

AMTA-218	OPTIMIZATION THEORY	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Basic Concepts: optimal problem formulation, single variable optimization algorithms: bracketing, region elimination, point estimation, and gradient based methods, root finding.

Multivariable optimization algorithms: unidirectional search, direct search methods, simplex search and gradient based methods.

Constrained optimization algorithms: penalty function method, method of multipliers, sensitivity analysis, and direct search for constrained minimization, linearized search techniques, feasible direction method, generalized reduced gradient method, and gradient projection method.

Non-traditional optimization algorithms: Genetic algorithms, simulated annealing, and global optimization. Computer programming practice for general design applications.

Texts / Reference Books:

1. Deb, K. Optimization for engineering design: algorithms and examples. Prentice Hall of India, New Delhi.
2. Rao, S.S.. Engineering Optimization: Theory and Practice. Wiley.
3. Bradley, S.; Hax, A.; Magnanti, T. Applied mathematical programming. Addison Wesley.
4. Rardin, Ronald L. Optimization in operations research. Prentice Hall.
5. Strang, Gilbert. Introduction to applied mathematics. Wellesley-Cambridge Press.

Course Description

Course Objectives

At the end of this course, students should:

1. Define the basics of simulation modeling and replicating the practical situations in organizations
2. Analysis of Simulation models using linearized search techniques, feasible direction method, and generalized reduced gradient method.
3. Explain non-traditional optimization algorithms.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-218.1	Describe the role of important elements of discrete event simulation and modeling paradigm
AMTA-218.2	Conceptualize real world situations related to multivariable optimization algorithms.
AMTA-218.3	Develop skills to apply optimization algorithms software to construct and execute goal-driven system models.
AMTA-218.4	Interpret the model and apply the results to resolve critical issues of optimal solution.

AMTA-219	MEMS AND NEMS	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Introduction to MEMS, MEMS sensors and actuators, scaling issues in MEMS, Micro and nano mechanics – principles, methods and strain analysis, Evolution of Micro sensors & MEMS, Micro sensors & MEMS applications, Introduction to Microelectronic technologies for MEMS.

Materials fabrication processes: Micromachining Technology, Surface and Bulk Micromachining, Micromachined Micro sensors, Mechanical, Inertial, Biological, Chemical, Acoustic, Microsystems Technology, Integrated Smart sensors and MEMS, Interface Electronics for MEMS, MEMS Simulators, MEMS for RF Applications, Bonding, Packaging and reliability packaging of MEMS, measurement techniques and future trends for MEMS.

Nanoelectromechanical systems (NEMS) – a journey from MEMS to NEMS, MEMS vs. NEMS, MEMS based nanotechnology – fabrication, film formation and micromachining.

NEMS physics – manifestation of charge discreteness, quantum electro dynamical (QED) forces, quantum entanglement and teleportation, quantum interference, quantum resonant tunnelling and quantum transport.

Electronic and photonic band gap crystals and their applications, NEMS architecture, Surface Plasmon effects and NEMS fabrication for nano photonics and Nano electronics.

References:

1. Electromechanical Sensors and Actuators, Ilene J. Busch-Vishniac, Springer, 2008
2. Introduction to Microelectronics Fabrication, Vol. V, G. W. Neudeck and R. F. Pierret (eds.), Addison, Wesley, 1988
3. Introduction to Microelectromechanical Microwave Systems, H. J. De Loss Santos, 2nd edition, Norwood, MA: Artech, 2004
4. Microsystems Design, S. D. Senturia, Kluwer – Academic Publishers, Boston MA, 2001.
5. Principles and Applications of Nano-MEMS Physics, H. J. Delos Santos, Springer, 2008.
6. Materials and Process Integration for MEMS Microsystems, Vol. 9, Francis E. H. Tay, Springer, 2002.
7. MEMS: Introduction and Fundamentals by Mohamed Gad-el-Hak –
8. MEMS mechanical sensors by Stephen Beeby
9. Microsensors, MEMS, and smart devices by Julian W. Gardner, V. K. Varadan, Osama O. Awadelkarim.
10. MEMS and microsystems: design, manufacture, and nanoscale engineering- Tai-Ran Hsu.
11. MEMS by N P Mahalik.

Course Description

Course Objectives

- At the end of this course, students should:
 1. Have a concept on the scope and recent development of the science and technology of micro- and nano-systems.
 2. Gain the physical knowledge underlying the operation principles and design of micro and nano-systems
 3. Learn some typical or potentially applicable micro- and nano-systems at the frontier of the development of the field

Course Outcome

Upon completion of their academic work student will be able:

AMTA-219.1	Ability to understand the operation of micro devices, micro systems and their applications
AMTA-219.2	Ability to design the micro devices, micro systems using the MEMS fabrication process.
AMTA-219.3	Gain a knowledge of basic approaches for various Nano electromechanical systems (NEMS).
AMTA-219.4	Gain a knowledge of basic approaches for quantum interference, quantum resonant tunnelling and quantum transport.
AMTA-219.5	Gain the technical knowledge required for NEMS architecture, and NEMS fabrication for nano photonics and nano electronic.

AMTA-220	DIGITAL MANUFACTURING	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Digital design: Geometrical design of curves, Surfaces and solids, Introduction to computer aided engineering analysis and optimum design. Consideration of manufacturing and assembly aspects in design.

Shape digitization: 3D object scanning, Solid reconstruction from point cloud and tessellated data, Downstream applications; Digital manufacturing: Subtractive manufacturing: Basic architecture, Control hardware and software details, Tooling, Sculptured surface machining;

Additive Manufacturing: Basics, Hardware details and capabilities of commercial systems, planning of material addition, Rapid tooling solutions;

Computer Aided Process Planning: CAPP and route sheet development, CAPP system, Computer aided plant layout, Computer Aided Production Planning and Control, Algorithms for CAPP;

Product Database Management Systems: Types, Management Information System, Manufacturing data preparation, Shop-floor control, automatic identification systems (sensors, trackers), Product life cycle management; and Introduction of Industry 4.0.

Texts / Reference Books:

1. Fundamentals of Digital Manufacturing Science, by Z.Zhou,S.Xie, D. Chen, Springer, 2012.
2. Rapid Prototyping: Principles and Applications By C.K. Chua, K.F. Leong, C.S. Lim, John Wiley, 2010.
3. Mastering CAD CAM By Ibrahim Zeid, McGraw Hill, 2005.
4. Automation, production systems, and computer-aided manufacturing By M P Groover, Pearson, 2016.

Course Description

Course Objectives

- At the end of this course, students should:
 1. This course aims to mobilize the engineering students on how the advancement of digital technologies including information integration can enhance factory performance from design stage to digital inventory stage.
 2. Understand challenges imposed by aggressive consumer needs of cost effective processing that can deliver efficient, flexible, and modular manufacturing of customized products using digital technologies and their data integrative.
 3. Can perform isolate basic operations, by applying integrated of information in digital form beginning from the resource model simulation of raw material, digital production operations until digital finished product model has been delivered.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-220.1	Understand the strategic difference between traditional factory and digital factory with Cyber Physical System.
AMTA-220.2	Apply information models of innovative ideas in physical hardware as well as software to real product. Technologies such as Additive Manufacturing (AM), Virtual Reality (VR)/ Augmented Reality (AR) can be practiced.
AMTA-220.3	Analyze works in both information-managed and file-based environments for the development of robotic and automated production systems.
AMTA-220.4	Robot and workstation simulation development, from single-robot stations to complete production lines and zones are used with collaborative tools in conjunction with data from sensors in digital based. Simulation software can centralize role of these analyzing information from sensor and mobile device to robotics and automated guided vehicle (AGV) for digital inventory.
AMTA-220.5	Design digital factory in virtual environment. Streamlining throughput using simulation software to design and validate of manufacturing processes in a dynamic and highly visual environment where the optimize manufacturing process sequences with realistic behaviour using fully kinematic 3D models can be performed.

AMTA-221	COMPUTER AIDED MANUFACTURING	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Basic principles of automation; extending the capabilities of conventional machines through improved devices and manipulators.

Basic principles of numerical control; CNC, DNC and Machining Centres.

Manual part programming (industrial controller like Fanuc/Siemens).

APT programming; System devices; Adaptive control; Economics of numerical control.

Introduction to Robotics: Synthesis of elements with movability constraints; Elements of robot anatomy; Hydraulic, pneumatic and electrical manipulators; End-effectors and their design; Controllers with microprocessors; Robot Sensors; Applications of industrial robots; Economics of robotics.

Texts / Reference Books:

1. CAD/CAM Theory and Practice, Ibrahim Zeid, Tata McGraw Hill Publication
2. CAD/CAM Principle and Application, P. N. Rao, Tata McGrawHill Publication.
3. Users guides to Rapid Prototyping, Todd Frimm, Society of Manufacturing Engineers
4. Computer control of Manufacturing system, Yoram koren, McGraw Hill Publication.
5. Machining and Metal Working Handbook, Ronal A Walsh and Denis Cormier McGraw Hill Publication.
6. Machining and CNC Technology, M. Fitzpatrick, McGraw-Hill Publication.
7. Journey from Rapid Prototyping to Rapid Manufacturing, S Chattopadhyaya, Lambert Publication.

Course Description

Course Objectives

- At the end of this course, students should:
 1. Apply contemporary knowledge to understand the principles of automation in industries.
 2. Analyze how to operate and control the different types of NC, CNC and DNC machine tools.
 3. To provide skills to write the manual part programing and APT programing to operate the machine tools.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-221.1	Understand basic principles of automation and capabilities of conventional machines.
AMTA-221.2	Study the basic principles of automation in machine tools .
AMTA-221.3	Understand the part programing and apply the manual part programming to control the various machining processes.
AMTA-221.4	Understand the APT part programing and apply the APT part programming to control the various machining processes.
AMTA-221.5	Basics of robotics and its elements. Applications of industrial robots; Economics of robotics

AMTA-222	INSPECTION AND TESTING IN MANUFACTURING	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Basics: Types and purposes of testing of manufactured components, Precautions in inspections, Accuracy of measurement and important terms.

Destructive Physical Analysis (DPA): Suitability and purpose; Review of Mechanical testing Methods. Tensile Testing (TT); Compression test, Charpy Impact test, Hardness Testing (HT) - Micro and Nano-hardness test, Stress Rupture Testing (SRT); Toughness, Fatigue and Fracture toughness test, Bend test, Creep test, Chemical tests, Macrographs study.

ASTM standard test methods: Tensile test, Charpy Impact test, Micro-hardness evaluation, Fracture toughness test, Crack growth rate study, Flexural strength of beam; Introduction to NDT, Visual Optical methods, Dye penetrant testing, Methods of application, Developer. Magnetic particle testing, Magnetization methods, Field indicators, Particle application, Inspection; Eddy current testing, Faraday's law, Inductance, Lenz's law, Self and Mutual Inductance, Impedance plane, Inspection system.

Ultrasonic testing: Basics of ultrasonic waves, Pulse and beam shapes, Ultrasonic transducers, Distance and Area calibration, Weld inspection by UT.

Acoustic emission testing: Sources of acoustic emission, Source parameters, Kaiser-Felicity theory, Equipment and Data analysis.

Radiography: X-rays and their properties, X-ray generation, X-ray absorption and atomic scattering. Image formation, Image quality, Digital Radiography, Image interpretation, Radiation Shielding.

ASTM standard test method for NDT tests, like Radiographic, Ultrasonic, Electromagnetic (eddy current), X-ray, Acoustic and Tomographic techniques; and Comparison and selection of NDT methods.

Texts / Reference Books:

1. Non-destructive Testing, Louis Cartz, ASM International
2. Non-destructive Evaluation and Quality Control, ASM Handbook, Vol. 17.
3. Non-Destructive Test and Evaluation of Materials By J Prasad, McGraw Hill, 2017
4. Welding Inspection, American Welding Society, 3rd Ed., 2000
5. The Mechanical Testing of Metals and Alloys By foster, P. Field, Cousens Press, 2007
6. Metals Handbook: Mechanical testing, American Society for Metals, 1978
7. ASTM standards for mechanical test, such as: ASTM E8/E8M (Tension test for metals), ASTM
8. D6110-10 (Charpy impact test), ASTM E9-09 (Compression test), ASTM E139-11 (Creep test)
9. ASTM standards for various non-destructive tests

Course Description

Course Objectives

- At the end of this course, students should:
 1. Understand quality control its used to manufacture products free from defects and meet customer requirements.
 2. Understand the importance of Inspection and measurement during production for quality control
 3. Familiar with the concept of non-destructive testing among the students and make them understand various types of non-traditional practices available for manufacturing industry.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-222.1	The student shall be able to select an appropriate NDT technique as per requirement.
AMTA-222.2	The student shall be able to set various process parameters and control the NDT process for the desired output parameters.
AMTA-222.3	The student shall be able to solve various problems encountered like leakage, cracks, blowholes etc. with the manufacturing process by analyzing the data.
AMTA-222.4	The student shall be competent enough to make use of modern tools and software's for analyzing and solving real life problems.
AMTA-222.5	The student shall be able to introduce environmental friendly solutions to achieve organizational sustainability.

AMTA-223	FLEXIBLE MANUFACTURING SYSTEM	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

FMS concept: Types of production, production planning and control, manufacturing in a competitive environment, concept, automation of manufacturing process, numerical control, adaptive control, material handling and movement, industrial robots, flexible fixturing, design for assembly, disassembly and service. types of FMS, types of FMS layouts, advantages and disadvantages of FMS Group technology – composite part families - classification and coding - production flow analysis, automated material handling system AGVs.

Automated process planning: Group Technology, Part families, Part classification and coding, Production flow analysis, Machine cell design, Applications and Benefits of Group Technology, Structure of a Process Planning, Process Planning function, CAPP - Methods of CAPP, CAD based Process Planning, **Inventory management:** Materials requirements planning - basics of JIT.

Monitoring and quality control: Types of production monitoring system, process control & strategies, direct digital control - Supervisory computer control - computer aided quality control - objectives of CAQC, QC and CIM, contact, non-contact inspection methods, CMM and Flexible Inspection systems. Integration of CAQC with CIM.

Texts / Reference Books:

1. Kant Vajpayee. S., “Principles of Computer Integrated Manufacturing”, Prentice Hall of India, 1999.
2. Radhakrishnan.P, Subramanyan. S, “CAD/CAM/CIM”, New Age International publishers, 2000.
3. Scheer.A.W., “CIM- Towards the factory of the future” Springer - Verlag, 1994.
4. Daniel Hunt.V., “Computer Integrated Manufacturing Hand Book”, Chapman & Hall, 1989.
5. Groover M.P, “Computer Aided Design and Manufacturing”, Prentice Hall of India, 1987.
6. Yorem Koren, “Computer Control of Manufacturing System”, McGraw Hill, 1986.
7. Ranky Paul. G., “Computer Integrated Manufacturing”, Prentice Hall International, 1986/81.
8. Automation Production System & CIM, M. P. Groover.
9. Materials & processes in Manufacturing, Degarmo, J. T. Black.
10. Lean Manufacturing System & Cell Design, J. T. Black.

Course Description

Course Objectives

The main objectives of the course are:

1. To introduce and discuss flexible manufacturing concepts;
2. To have the students gain insight about the state-of-the-art research areas related to FMS and real-time shop floor control
3. To have the students gain advanced skills in modeling, design and simulation of complex systems

Program Outcomes:

Course Outcome

Upon completion of their academic work student will be able:

AMTA-223.1	Apply the concepts of PPC and GT to the development of FMS.
AMTA-223.2	Identify various workstations, system support equipment's.
AMTA-223.3	Discuss the planning and scheduling methods used in manufacturing systems.
AMTA-223.4	Identify hardware and software components of FMS.
AMTA-223.5	Summarize the concepts of modern manufacturing such as JIT, supply chain management and lean manufacturing etc

AMTA-224	MODELLING AND SIMULATION	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Physical Modelling: Mechanical and electrical systems, physical laws, continuity equations, compatibility equations, system engineering concept, system modelling with structured analysis, modelling paradigms for mechatronic system, block diagrams, mathematical models, systems of differential-algebraic equations, response analysis of electrical systems, thermal systems, fluid systems, mechanical rotational system, electrical-mechanical coupling.

Simulation Techniques: Solution of model equations and their interpretation, zeroth, first and second order system, solution of 2nd order electro-mechanical equation by finite element method, transfer function and frequency response, non-parametric methods, transient, correlation, frequency, Fourier and spectra analysis, design of identification experiments, choice of model structure, scaling, numeric methods, validation, methods of lumped element simulation, modelling of sensors and actuators, hardware in the loop simulation (HIL), rapid controller prototyping, coupling of simulation tools, simulation of systems in software (MATLAB, LabVIEW) environment.

Texts / Reference Books:

1. L. Ljung, T. Glad, “Modeling of Dynamical Systems”, Prentice Hall Inc. (1994).
2. D.C. Karnopp, D.L. Margolis and R.C. Rosenberg, “System Dynamics: A Unified Approach”, 2nd Edition, Wiley-Interscience (1990).
3. G. Gordon, “System Simulation”, 2nd Edition, PHI Learning (2009).
4. V. Giurgiutiu and S. E. Lyshevski, “Micromechatronics, Modeling, Analysis, and Design with MATLAB”, 2nd Edition, CRC Press (2009).

Course Description

Course Objectives

- At the end of this course, students should:
 1. Students will understand the techniques of modeling in the context of hierarchy of knowledge about a system and develop the capability to apply the same to study systems through available software
 2. Students will learn different types of simulation techniques.
 3. Students will learn to simulate the models for the purpose of optimum control by using software

Course Outcome

Upon completion of their academic work student will be able:

AMTA-224.1	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
AMTA-224.2	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
AMTA-224.3	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
AMTA-224.4	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
AMTA-224.5	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

AMTA-225	COMPUTATIONAL FLUID DYNAMICS	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Concept of Computational Fluid Dynamics: Different techniques of solving fluid dynamics problems, their merits and demerits, governing equations of fluid dynamics and boundary conditions, classification of partial differential equations and their physical behavior, Navier-Stokes equations for Newtonian fluid flow, computational fluid dynamics (CFD) techniques, different steps in CFD techniques, criteria and essentialities of good CFD techniques.

Finite Difference Method (FDM): Application of FDM to model problems, steady and unsteady problems, implicit and explicit approaches, errors and stability analysis, direct and iterative solvers.

Finite Volume Method (FVM): FVM for diffusion, convection-diffusion problem, different discretization schemes, FVM for unsteady problems.

Prediction of Viscous Flows: Pressure Poisson and pressure correction methods for solving Navier-Stokes equation, SIMPLE family FVM for solving Navier-Stokes equation, modelling turbulence.

CFD for Complex Geometry: Structured and unstructured, uniform and non-uniform grids, different techniques of grid generations, curvilinear grid and transformed equations. Lattice Boltzman.

Molecular Dynamics: Boltzman equation, Lattice Boltzman equation, Lattice Boltzman methods for turbulence and multiphase flows, Molecular interaction, potential and force calculation, introduction to Molecular Dynamics algorithms.

Texts / Reference Books:

1. J. D. Anderson, "Computational Fluid Dynamics", McGraw-Hill Inc. (1995).
2. S. V. Patankar, "Numerical Heat Transfer and Fluid Flow", Hemisphere Pub. (1980).
3. K. Muralidhar, and T. Sundarajan, "Computational Fluid Flow and Heat Transfer", Narosa (2003).
4. D. A. Anderson, J. C. Tannehill and R. H. Pletcher, "Computational Fluid Mechanics and Heat Transfer", Hemisphere Pub. (1984).
5. M. Peric and J. H. Ferziger, "Computational Methods for Fluid Dynamics", Springer (2001).
6. H. K. Versteeg and W. Malalaskera, "An Introduction to Computational Fluid Dynamics", Dorling Kindersley (India) Pvt. Ltd. (2008).
7. C. Hirsch, "Numerical Computation of Internal and External Flows", Butterworth-Heinemann, (2007).
8. J. M. Jaile, "Molecular Dynamics Simulation: Elementary Methods", Willey Professional, 1997.
9. A. A. Mohamad, "Lattice Boltzman Method: Fundamentals and Engineering Applications with Computer Codes", Springer (2011).

Course Description

Course Objectives

- At the end of this course, students should:
 1. Apply contemporary knowledge to understand the different computational fluid dynamics techniques to solve the fluid dynamics problems.
 2. Analyze and develop the models to for steady and unsteady problems, implicit and explicit approaches and convection-diffusion to solve the problems.
 3. To provide skills to model and simulate the any fluid mechanics related problem for complex geometries.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-225.1	Basic understanding of computational fluid dynamics its fundamental governing equations and different solving techniques.
AMTA-225.2	Different approaches to solve the problems such as finite difference method and finite volume methods.
AMTA-225.3	Prediction and calculation of different constants and parameters for solving navier-stokes equation.
AMTA-225.4	Apply the computational fluid dynamics approach on complex geometries under conditions.
AMTA-225.5	Basics of molecular dynamics algorithms and calculation of potential and force using boltzman equation, lattice boltzman equation.

AMTA-226	DRONE TECHNOLOGY	L 3	T 0	P 0	C 3
Prerequisite	NIL				

Syllabus:

Drones: Unmanned Aircraft Systems, History, Classification, System Composition, Aerodynamics. Fundamentals of flight, DGCA safety regulations.

Drone's mechanical parts: Design and Selection of the System, Identification and selection of different components, parts, block of the drone and its function & their interconnectivity, Recognize basic principles of flying like Bernoulli's Principle. Types of propeller designs used in drones.

Drone's electronic parts: Different electronic components and their measurement, test different sensors, Identify, test and install different sensors on drone. Type of batteries, specifications and their charging techniques. DC, BLDC and servo motors and test driving circuits. Drone Electronic Speed Controller (ESC).

Communication, Payloads and Controls: Inspect, test and execute GPS navigation and telemetry module, different RF blocks and antennas used in RF transmitter and receiver. Payloads, controls-PID feedback, radio control frequency range, modems, memory system, simulation, ground test-analysis-trouble shooting.

Navigation, Testing and Drone pilot training: Waypoints navigation, ground control software, System Ground Testing, System In-flight Testing, Calibrate and troubleshoot drone gimbal and drone payload, Identify and resolve common error messages and corrections by software debugging, Drone equipment fabrication, Future Prospects and Challenges.

References and Text Books:

1. Reg Austin "Unmanned Aircraft Systems UAV design, development and deployment", Wiley, 2010.
2. Robert C. Nelson, Flight Stability and Automatic Control, McGraw-Hill, Inc, 1998.
3. Kimon P. Valavanis, "Advances in Unmanned Aerial Vehicles: State of the Art and the Road to Autonomy", Springer, 2007
4. Paul G Fahlstrom, Thomas J Gleason, "Introduction to UAV Systems", UAV Systems, Inc, 1998.
5. Dr. Armand J. Chaput, "Design of Unmanned Air Vehicle Systems", Lockheed Martin Aeronautics.
6. https://dgt.gov.in/sites/default/files/DRONE%20_TECHNICIAN%20_NSQF%20_LEVEL_4.pdf.

Course Description

Course Objectives

- At the end of this course, students should:
 1. . Apply contemporary knowledge to understand the fundamental, history and classification of drone technologies.
 2. Design and selection of mechanical and electronics components of a drone for different applications.
 3. To provide skills to control and communication of drones. Provide the training and testing of flying the drones.

Course Outcome

Upon completion of their academic work student will be able:

AMTR-226.1	Understand basic concepts of Drones and its types.
AMTR-226.2	Apply navigation concepts in drone's pilot training.
AMTR-226.3	Analyze communication, pay loads and controls in drones.
AMTR-226.4	Design and selection of electronic and mechanical parts of drones.
AMTR-226.5	Assess future prospects and challenges in field of drone fabrication.

AMTA-251	ROBOTICS AND AUTOMATION LAB	L 0	T 0	P 2	C 1
Prerequisite	NIL				

List of Experiments:

1. Introduction to 6 Axis KUKA robots.
2. Jogging of the 6 Axis KUKA robots.
3. Manual Motion of End Effector for point P1 to point P2 by Smartpad (Jogging Pad)
4. Pre-program motion of end effector from point T1 to T2.
5. Program motion of end effector from point T1 to T2 in looping manner
6. Study of operation of an ASRS
7. Study the operation of Lathe machine
8. Study the operation of AGV

Course Objectives

At the end of this course, students should:

1. Lab allows students to design, test and build autonomous and automated systems that are used in modern manufacturing.
2. Teach the fundamentals of manipulator kinematics, dynamics, control and programming with a close link to the real robots used in industry.
3. Practical implementation of the theoretical understanding of robotic systems.

Course Outcome:

Upon completion of their academic work student will be able:

AMTA-251.1	The students can interact directly with devices such as industry standard sensors and actuators through Human Machine Interfaces (HMIs) and PCs. Gather, process and monitor experimental data locally or at remote stations/locations.
AMTA-251.2	Student can learn advanced controllers and their applications in industry.
AMTA-251.3	Understand the relevance of teaching material to the real world systems that are currently used in industry, which will ultimately enhance their employability.
AMTA-251.4	Understand the basic process for programming the industrial manipulator. This relies on understanding the different links, frames or coordinate systems of the manipulator.
AMTA-251.5	Explain Cartesian controllers of the manipulator, using the forward and inverse kinematics and dynamics of the manipulator. Understand joint controller in the Robot Operating System. Setting each joint of the arm to the values specified by the inverse kinematic model, and allowing them to specify arbitrary positions and orientation in the world, and driving the manipulator to specified position.

AMTA-252	Independent Study & Research-II (ISR-II)	L 0	T 0	P 3	C 1
Prerequisite	NIL				

In this course, student will do a variety of activities.

1. The student will work on the problem identified in ISR1 and to prepare a manuscript based on the works done.
2. The students are encouraged for publication in SCI/Scopus indexed journals and conferences.
3. This course will provide enough wisdom and idea to the students for starting and selecting a good problem for the thesis dissertation.
4. Progress will be checked once a month for support and direction. A mentor will be assigned to each student for supervision.

Course Objective

The main objectives of course are:

1. The student will learn to work on research problems.
2. The basic purpose of this lab is to prepare students for good research works.
3. One of the objectives is also to help students in selection of their thesis dissertation topic.

Course Outcome

Upon completion of their academic work student will be able:

AMTA -252.1	Understand basic concepts of research.
AMTA -252.2	Apply previous literature review in selection of a workable research problem.
AMTA -252.3	Analyze the methods through which a research problem can be solved.
AMTA -252.4	Design a scheme to do work on research problem and to achieve targets.
AMTA -252.5	Assess challenges in doing research work based on literature reviews.

SEMESTER-III
CORE COURSES

AMTA-301	RESEARCH METHODOLOGY AND TECHNICAL WRITING	L 3	T 0	P 0	C 3
Prerequisite	None				

Syllabus:

Research Methodology:

Research process, types of research, problem identification and hypothesis formulation, Research design, methods of data collection, reliability and validity, data presentation, and report preparation.

Introduction to Research Communication:

Grammar and Rhetoric: Sentential and supra sentential structure, Narrative and structuring argument, common error in composition. Reading skills for literature review: Previewing techniques, understanding the gist of an argument, identifying the topic sentence.

Writing skills (Part-I)

Sentence formation, Use of appropriate diction, paragraph and essay writing, coherence and cohesion. Summarizing, paraphrasing, outlining, Non-linear description, Narrative, Instruction and reporting. Descriptive and explanatory, analytical and argumentative writing, enhancing editing skills, punctuation.

Writing skills (Part-II)

Introduction to terminology, concept of research. Preparing research proposal/ Synopsis. Formulating thesis statement. Referencing (all style sheet). Writing Introduction, Footnotes/ Endnotes, Conclusion. Preparing Appendix, Bibliography (all style sheets), and Abstract.

Writing acknowledgement. Concept of Keywords, preparing content page/ list of Tables and Figures. Use of classified materials, Plagiarism and copyright materials.

Texts / Reference Books:

1. Ranjit Kumar, research methodology: a step-by-step guide for beginners, SAGE Publications India Pvt Ltd.
2. C.R. Kothari, Research Methodology: Methods and Techniques, New Age International, 2004.
3. Michael H. Markel. Handbook of technical writing. Bedford/St Martins, 2012
4. Technical Writing and Professional Communication: For Nonnative Speakers of English. Thomas N. Huckin. McGraw Hill 1991.
5. Michael H. Markel. Technical Writing Essentials. St. Martin's Press, 01-May-1988.
6. Sharon J Gerson, Steven M Gerson. Technical writing: Process and product. Fifth edition, Pearson Education.

Course Description

Course Objectives

At the end of this course, students should:

1. Develop skills in the use of tools and techniques required for conducting qualitative research.
2. Plan, conduct, present and discuss qualitative research study.
3. Apply the skills to writing proposals and research/project reports.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-301.1	Independently design and conduct a qualitative research regarding methods of data collection, data presentation, and report preparation.
AMTA-301.2	Describe, and apply qualitative research tools like research communications to explore the reading skills for literature review interviewing.
AMTA-301.3	Understand the skills of sentence formation, use of appropriate diction, paragraph, summarizing, paraphrasing, and outlining,
AMTA-301.4	Write preparing research proposal/ synopsis, formulating thesis statement and referencing (all style sheet).

AMTA-351	THESIS / DISSERTATION - I	L 0	T 0	P 14	C 9
Prerequisite	None				

Course Contents

The student will be encouraged to finalize the area of the project work during the end of second semester itself. The project works related to industry specific problem solving are also encouraged. The project can be carried out in the University or in collaboration with an industry/research organization/other University. If a student undergoes his/her project work outside the University, one External Supervisor from the organization will be there along with one faculty member from the Department as an Internal Supervisor. In the end of the third semester, student has to present a seminar on the progress of his/her research work. A brief project report needs to be submitted during the presentation of the work. There shall be evaluation of the work carried out at the end of the third semester.

Course Description

Course Objectives

At the end of this course, students should:

1. To provide students with opportunity to learn new concepts and skills acquired in core courses and further extend these ideas to solve research/industry related problems.
2. The students will identify a research area of interest and a relevant research problem to work on.
3. A list of objectives and deliverables along with the timeline are also identified.
4. Research Progress Seminar will be held twice a month for continuous evaluation.
5. The students will continue to work on the problem identified in Major Project (in Semester IV) also.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-351.1	Explain factual knowledge (terminology, classifications, methods, trends) of current areas of research.
AMTA-351.2	Understand the steps for systematic research and the parameters involved in the research process.
AMTA-351.3	State and explain some fundamental principles, generalizations, or theories to solve the research problem.
AMTA-351.4	Complete implementation of an appropriate research design by achieving the specific skills and competencies to draw appropriate conclusions from research work.
AMTA-351.5	To judge the value of different contributions and identify promising new directions of the work. Submit the report regarding conference participation or Journal publication with experimental result, result analysis.

SEMESTER-IV

AMTA-451	THESIS/DISSERTATION - II	L 0	T 0	P 26	C 13
Prerequisite	None				

Course content

The students will continue to work on the problem identified in “Thesis/Dissertation-I” (in Semester III) as per the work plan. The work is continued until all stated objectives and deliverables are met. Student will prepare a comprehensive report containing introduction to the problem, literature review, methodology, results and discussion and conclusion. On completion of the project work, the student shall submit a thesis to the department for examination. There shall be evaluations of the project work by a committee constituted by the department with an external examiner and candidate has to appear an open viva-voce examination on his/her thesis. Research Progress Seminar will be held twice a month for continuous evaluation.

Course Description

Course Objectives

At the end of this course, students should:

6. Design of the formulated solution and develop skills in the use of tools and techniques required for conducting qualitative research.
7. Plan and conduct experimental procedures, test procedures, data collection.
8. Analysis of the results and apply the skills to writing proposals and research/project reports.

Course Outcome

Upon completion of their academic work student will be able:

AMTA-451.1	Formulate a research problem relevant to society/industry through critical review of appropriate and relevant information sources.
AMTA-451.2	Systematically identify relevant theory towards a research related problem, relate these to appropriate methodologies and evidence and perform experiments/modelling, etc.
AMTA-451.3	Apply qualitative and/or quantitative evaluation method to obtain the outcome of the research work undertaken.
AMTA-451.4	Complete implementation of an appropriate research design and associated methods rigorously and draw appropriate conclusions from research work.
AMTA-451.5	Prepare summary of research work in a coherent and convincing way by providing concrete findings and key arguments in the form of thesis report. Submit the report regarding conference participation or Journal publication with experimental result, result analysis.